

JUL-SINTECE Integration Control Document



CNCA Certificate Issuance Process

Amendment and Cancellation Capabilities

Version 3.0
Abu Dhabi Ports
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 Document Control

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Version	Date	Author	Description of Changes
1.0	2025-11-12	Linoy Pappachan Malakkaran	Initial ICD creation with technical specifications for JUL-SINTECE integration. Defined API specifications, data models, integration workflows, security requirements, and testing procedures.
2.0	2025-11-13	Linoy Pappachan Malakkaran	Added amendment and cancellation APIs, approval workflows, status management with canAmend/canCancel flags, validation framework, and error handling. Added process flow improvements and implementation recommendations.
3.0	2025-11-14	Linoy Pappachan Malakkaran	Removed flow diagrams, webhook references, payment processing references. Added Keycloak SSO authentication, NIF validation API, invoice download API, status retrieval API, and consolidated CNCA certificate submission API. Updated table format to 6 columns.



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1. Introduction

1.1 Purpose

This Interface Control Document (ICD) defines the technical specifications and interface requirements for integrating the JUL system (Abu Dhabi Ports) with the SINTECE system (ARCCLA) for CNCA certificate issuance.

Key Features:

- Certificate submission and approval workflow
- Amendment and cancellation requests
- Master data synchronization
- Status tracking and notifications

1.2 Scope

In Scope:

- Certificate request lifecycle (creation to issuance)
- Amendment and cancellation workflows
- API specifications and data models
- Authentication and authorization
- Validation and error handling

Out of Scope:

- Internal SINTECE business logic
- Payment gateway implementation
- Mobile application development
- Legacy system migration

1.3 Audience

- **Developers:** API implementation and integration
- **Architects:** System design and technical decisions
- **QA Engineers:** Testing and validation
- **Business Analysts:** Requirements validation
- **Project Managers:** Implementation tracking

1.4 Key Terms

Term	Definition
ARCCLA	Angolan regulatory agency responsible for cargo certification
BL	Bill of Lading - shipping document required for certificates
CNCA	Certificado Nacional de Carga de Angola - required certificate for Angola cargo
CTN	Certificate/Cargo Tracking Note - internal certificate reference
DUP	Declaration of Unique Property - unique customs identifier
JUL	Abu Dhabi Ports system for CNCA certificate management

SINTECE	ARCCLA system for certificate processing and approval
Amendment	Request to modify submitted certificate data
Cancellation	Request to cancel submitted certificate
canAmend/canCancel	System flags indicating eligibility for amendment/cancellation

2. System Overview

2.1 System Architecture

JUL System (Abu Dhabi Ports)

- Trader portal for CNCA certificate management
- User interface for request creation and document upload
- Amendment/cancellation request capabilities
- Certificate download and status tracking
- Real-time status tracking and notifications

SINTECE System (ARCCLA)

- Backend certificate processing and approval system
- ARCCLA broker review workflow
- Amendment/cancellation approval processes
- Certificate issuance and status management
- Master data management and reporting

Integration Layer

- RESTful APIs with JSON data format
- JWT authentication and secure communication
- Real-time API calls and status notifications
- Error handling and audit logging

2.2 Integration Pattern

Communication:

- RESTful APIs with standard HTTP methods (GET, POST, PUT, DELETE)
- JSON data format with schema validation
- HTTPS/TLS encryption and JWT authentication
- OData query support for filtering and pagination

Data Flow:

1. **Initial CTN Creation:** JUL creates basic CTN record with core information via POST /api/ctns
2. **Section-wise Data Addition:** JUL adds detailed information in separate steps (internal to JUL system):
 - **Addresses:** Shipper, consignee, forwarder, notify party details
 - **Goods:** Cargo descriptions, classifications, weights, values
 - **Containers:** Container types, numbers, seals
 - **Tracking:** Transport routes, vessels, ports, schedules
 - **Attachments:** Supporting documents
3. **Certificate Validation:** SINTECE validates complete certificate data with business rules
4. **Visa Request Submission:** JUL submits for approval via POST /api/ctns/actions/requestvisa/{id}
5. **ARCCLA Review:** ARCCLA broker reviews and approves/rejects certificate
6. **Certificate Issuance:** Certificate issued with amendment/cancellation capabilities
7. **Status Updates:** Status communicated via API polling and notifications

Amendment Flow:

1. JUL validates amendment eligibility using canAmend status flag
2. JUL submits amendment request via POST /api/CTNs/{id}/amendments with change tracking

3. SINTECE validates amendment request and notifies ARCCLA broker
4. ARCCLA broker reviews amendment in SINTECE with impact analysis
5. SINTECE sends amendment approval/rejection to JUL via notifications
6. If approved, SINTECE updates certificate and certificate status
7. JUL updates certificate status and notifies stakeholders

Cancellation Flow:

1. JUL validates cancellation eligibility using canCancel status flag
2. JUL submits cancellation request via POST /api/CTNs/{id}/cancellations with reason documentation
3. SINTECE validates cancellation request and calculates financial impact
4. ARCCLA broker reviews cancellation in SINTECE with impact analysis
5. SINTECE sends cancellation approval/rejection to JUL via notifications
6. If approved, SINTECE processes cancellation and updates certificate status

2.3 System Actors

The following actors interact with the integrated system:

Actor	Role	System Access
Trader (Importer/Exporter)	Business entity shipping goods to/from Angola	JUL System (Web Portal)
Customs Broker / Freight Forwarder	Licensed agent representing trader with amendment/cancellation capabilities	JUL System (Web Portal with Amendment/Cancellation Features)
ARCCLA Broker	Government official authorized to approve certificates, amendments, and cancellations	SINTECE System (Internal Portal with Amendment/Cancellation Workflows)

Enhanced User Interaction Flow:

Standard Certificate Flow:

1. **Trader** logs into enhanced JUL → Creates certificate request → Uploads documents → Nominates broker
2. **Customs Broker** logs into enhanced JUL → Accepts nomination → Completes application with real-time validation → Submits to SINTECE
3. **ARCCLA Broker** logs into enhanced SINTECE → Reviews request with decision support tools → Approves/Rejects → Generates certificate (if approved)
4. **Trader** and **Customs Broker** receive certificate in JUL → Download and use for customs clearance

Amendment Flow:

1. **Customs Broker** checks amendment eligibility in JUL → Creates amendment request → Documents changes
2. **ARCCLA Broker** receives amendment notification in SINTECE → Reviews changes with impact analysis → Approves/Rejects
3. **Customs Broker** receives amendment decision in JUL → Updates stakeholders
4. **Trader** receives updated certificate with amendment history in JUL

Cancellation Flow:

1. **Customs Broker** checks cancellation eligibility in JUL → Creates cancellation request → Documents reasons
2. **ARCCLA Broker** receives cancellation notification in SINTECE → Reviews with impact analysis → Approves/Rejects
3. **Customs Broker** receives cancellation decision in JUL → Updates stakeholders
4. **Trader** receives cancellation confirmation

2.4 Amendment and Cancellation Capabilities

Amendment Management:

- Real-time eligibility validation based on certificate status
- Change tracking with before/after value comparison
- Impact analysis for proposed amendments
- Role-based approval workflow
- Complete audit trail with history

Cancellation Management:

- Real-time eligibility validation with financial impact assessment
- Reason code documentation
- Role-based approval workflow with escalation
- Complete audit trail

Enhanced Status Management:

- Dynamic eligibility flags (canAmend, canCancel) based on real-time business rules
- Enhanced status workflow supporting amendment/cancellation states
- Real-time status tracking and notifications
- Comprehensive status history with audit trail
- Status-based UI enhancements for improved user experience

Advanced Validation Framework:

- Multi-level validation (field-level, business rule, cross-system)
 - Real-time validation with immediate feedback
 - Configurable validation rules with role-based exceptions
 - Comprehensive error messaging with corrective action guidance
 - Integration testing validation for amendment/cancellation workflows
-

3. Process Flows

3.1 As-Is Process Analysis

The current SINTECE workflow consists of 15 distinct steps (LC-AR-CNCA-01 through LC-AR-CNCA-15) that involve multiple manual handoffs between traders, customs brokers, and ARCCLA staff.

Process Inefficiencies:

- Multiple email-based communications causing delays
- Manual document handling and verification processes
- Limited real-time visibility for stakeholders
- Duplicate data entry requirements across systems
- No automated amendment or cancellation capabilities

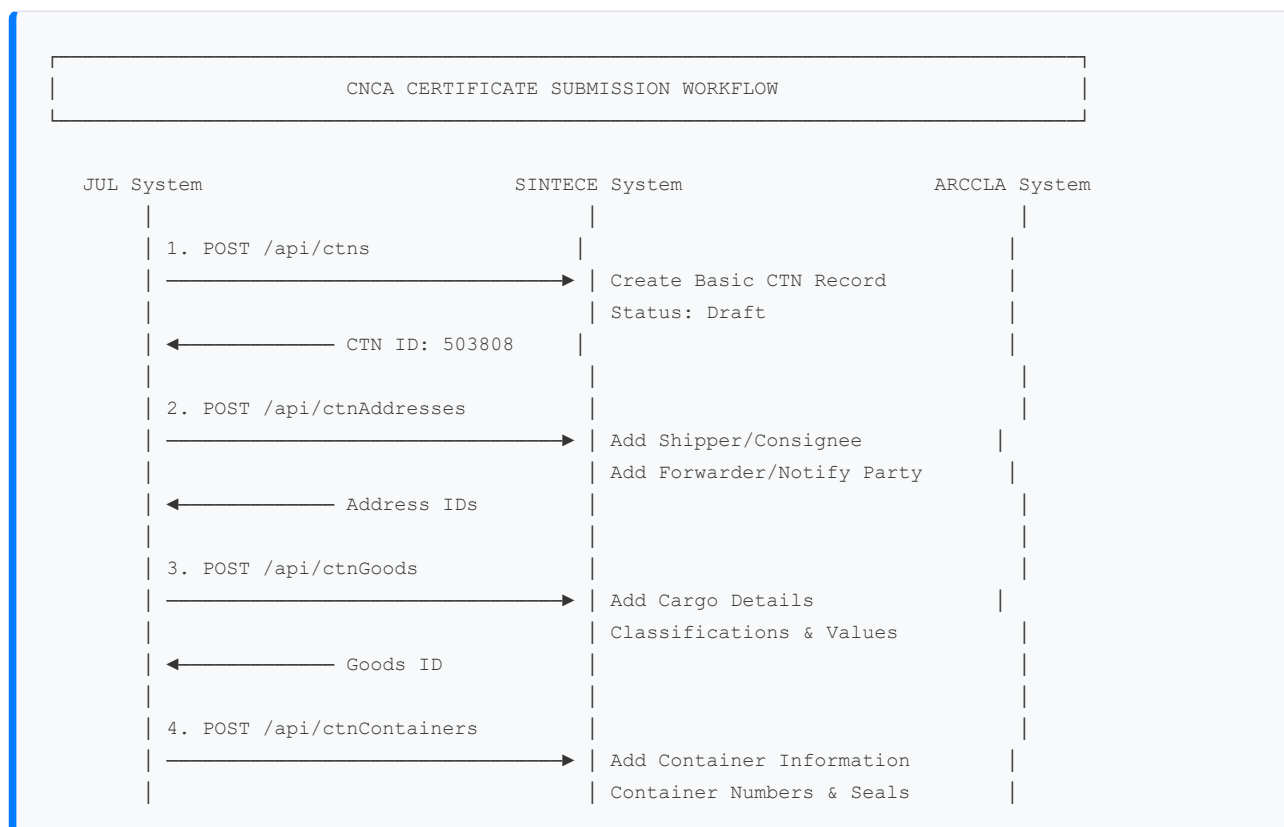
Key Bottlenecks:

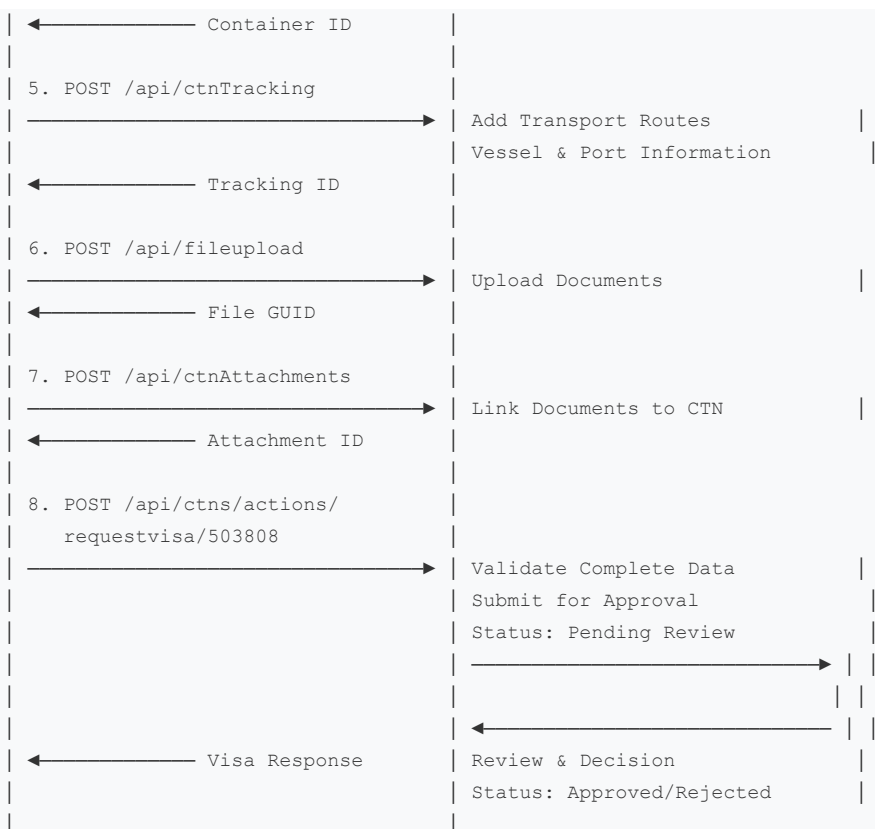
- ARCCLA broker manual review and approval steps
- Payment processing and reconciliation delays
- Physical document submission requirements
- Manual invoice generation and distribution

3.2 To-Be Process

The JUL-SINTECE integration introduces a fully digital workflow with section-by-section certificate submission capabilities to improve data accuracy and user experience.

3.2.1 Section-by-Section Certificate Submission Workflow





SECTION-BY-SECTION BENEFITS:

- ✓ Improved User Experience - Step-by-step data entry
- ✓ Enhanced Data Quality - Focused validation per section
- ✓ Progress Tracking - Users can save and resume
- ✓ Flexible Workflow - Optional sections based on cargo type
- ✓ Error Isolation - Issues contained to specific sections
- ✓ API Efficiency - Smaller, targeted API calls

Core Certificate Process Improvements:

1. **Digital Initiation:** Traders initiate requests directly through JUL system with section-by-section data entry
2. **Progressive Document Building:** Step-by-step completion of certificate sections for improved data quality
3. **Automated Validation:** Real-time validation of submitted data at each section
4. **Streamlined Approval:** ARCCLA brokers receive complete, validated requests for review
5. **Integrated Payment:** Seamless payment processing with automatic reconciliation
6. **Digital Certificate Issuance:** Electronic certificate generation and delivery

Enhanced Capabilities Integration:

- Real-time amendment requests with ARCCLA approval workflow
- Cancellation management with refund processing capabilities
- Status polling with eligibility flags for allowed operations
- Comprehensive audit trail and notifications system

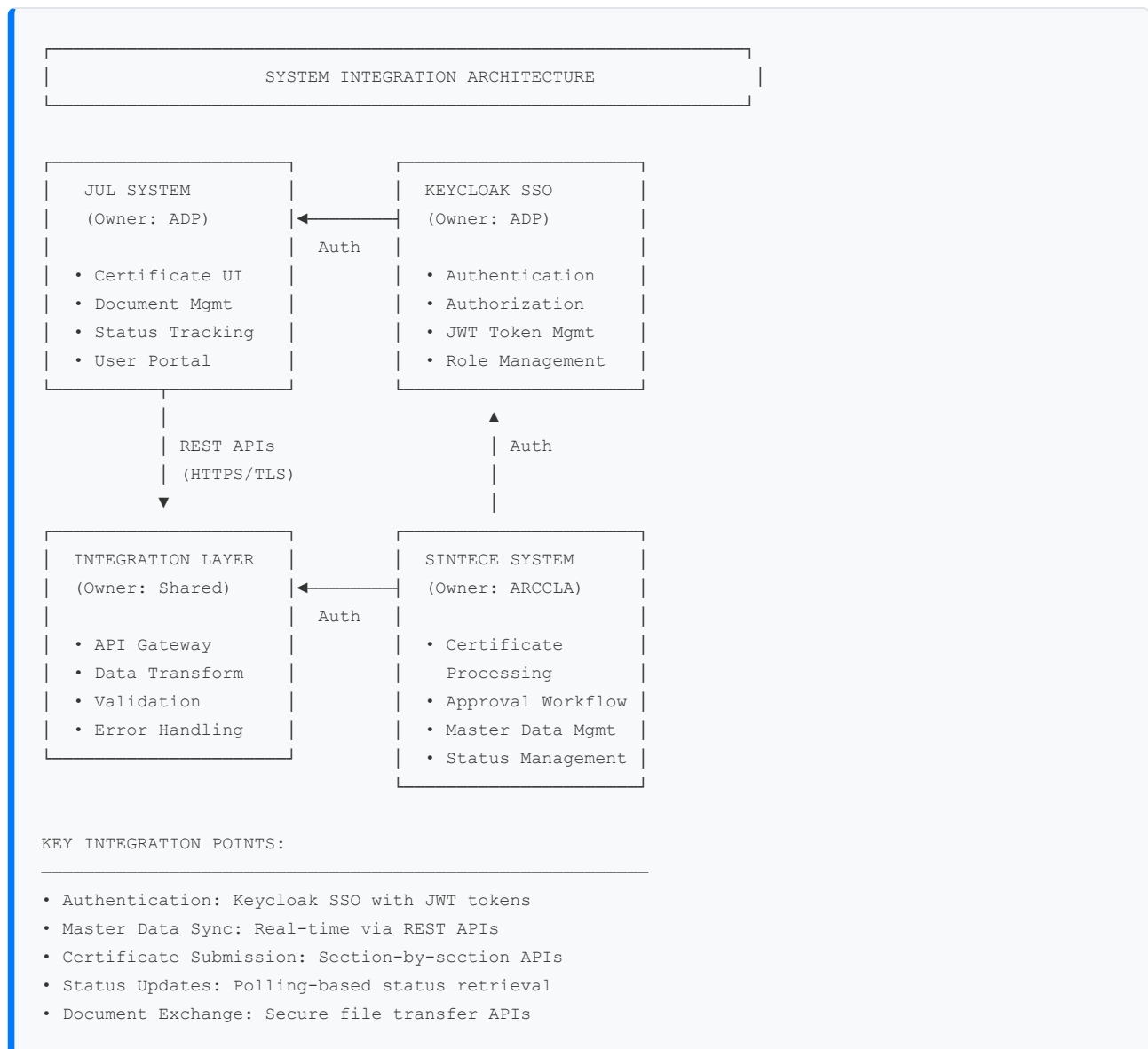
Benefits of Enhanced Process:

- Fully digital workflow with no physical documents
- Real-time status visibility for all parties
- Integrated amendment and cancellation capabilities
- Automated payment processing
- Single data entry point
- Complete audit trail

3.3 System Integration Architecture

The integration establishes comprehensive touchpoints between JUL and SINTECE systems with a modern API-driven architecture and centralized authentication.

System Integration Overview:



Component Ownership Table:

Component	Owner	Responsibilities

JUL System	Abu Dhabi Ports (ADP)	User interface, certificate request initiation, document upload, status display, user notifications, certificate download
Keycloak SSO	Abu Dhabi Ports (ADP)	User authentication, authorization, JWT token issuance and validation, session management, role-based access control
Integration Layer	Shared (ADP + ARCCLA)	API gateway, request/response transformation, data validation, error handling, logging and monitoring
SINTECE System	ARCCLA (Angola)	Certificate processing, approval workflows, master data management, certificate issuance, status management

Integration Flow Description:

1. User authenticates via Keycloak SSO (OpenID Connect/OAuth2)
2. JUL obtains JWT access token for API authorization
3. JUL sends certificate data to SINTECE via REST APIs
4. Integration layer validates and transforms requests
5. SINTECE processes certificate and updates status
6. JUL polls SINTECE for status updates
7. Certificate issued and available for download

Data Synchronization Points:

- Master data synchronization (countries, ports, cargo types)
- User account and role management coordination
- Certificate status and lifecycle management
- Amendment and cancellation request handling

API Integration Points:

- Authentication and authorization endpoints
- Certificate submission and retrieval APIs
- Amendment request creation and approval APIs
- Cancellation request creation and approval APIs
- Status monitoring and notification APIs

Business Process Touchpoints:

- ARCCLA broker approval workflows
- Document management and storage
- Audit logging and compliance reporting

3.4 Amendment Workflow Process

The amendment workflow provides controlled modification capabilities for issued certificates with comprehensive ARCCLA approval processes.

Amendment Request Initiation:

1. **Eligibility Check:** System validates certificate status and amendment permissions
2. **Request Creation:** Customs broker selects fields to amend and provides change reasons
3. **Validation:** System performs business rule validation on proposed changes
4. **Submission:** Amendment request is submitted to ARCCLA for review

ARCCLA Review and Approval:

1. **Notification:** ARCCLA broker receives amendment request notification
2. **Impact Analysis:** Review of proposed changes and potential impacts
3. **Decision:** Approval or rejection with detailed reasoning
4. **Processing:** For approved amendments, certificate is updated and new invoice generated

Amendment Eligibility Rules:

- Certificate status must be "Submitted", "Approved", or "Issued"
- No pending amendments or cancellations exist
- Certificate not yet used for customs clearance
- Amendment requested within allowed timeframe (typically 24 hours after issuance)
- User has appropriate amendment permissions

3.5 Cancellation Workflow Process

The cancellation workflow provides controlled termination of certificate requests with comprehensive financial impact management and refund processing.

Cancellation Request Initiation:

1. **Eligibility Check:** System validates certificate status and cancellation permissions
2. **Impact Assessment:** System calculates impact and eligibility
3. **Request Creation:** Customs broker provides cancellation reason and additional details
4. **Submission:** Cancellation request is submitted to ARCCLA for review

ARCCLA Review and Processing:

1. **Notification:** ARCCLA broker receives cancellation request notification
2. **Review:** Validation of cancellation reason and business justification
3. **Analysis:** Review of cancellation impact and timing
4. **Decision:** Approval or rejection of cancellation request
5. **Processing:** For approved cancellations, certificate is cancelled

Cancellation Eligibility Rules:

- Certificate status allows cancellation (not yet issued or used for customs clearance)
- Cancellation requested within allowed timeframe
- No active amendments in progress
- Valid business reason for cancellation provided

3.6 ARCCLA Approval Workflows

The ARCCLA approval workflows are enhanced to support the new amendment and cancellation capabilities:

Enhanced Review Dashboard:

- Unified queue for certificate submissions, amendments, and cancellations
- Priority-based processing with configurable business rules
- Rich context information for informed decision making
- Integrated communication tools for broker-to-broker coordination

Approval Decision Framework:

- Structured decision criteria for different request types
- Automated recommendations based on historical patterns
- Escalation procedures for complex or high-value requests
- Audit trail capture for all approval decisions

Workflow Automation:

- Automatic assignment of requests to available brokers
- SLA monitoring with escalation triggers
- Batch processing capabilities for high-volume periods
- Integration with existing SINTECE approval processes

Enhanced Notification System:

- Real-time notifications to all stakeholders
 - Email and system notifications with rich content
 - Configurable notification preferences by user role
-

4. Data Models

4.1 Core CTN Entity Structure

The Certificate Tracking Note (CTN) entity serves as the foundation for all certificate operations. The enhanced data model includes additional fields to support amendment and cancellation workflows:

```
{
  "id": "string (UUID)",
  "certificateNumber": "string (auto-generated)",
  "requestDate": "datetime (ISO 8601)",
  "submissionDate": "datetime (ISO 8601)",
  "status": "string (enum)",
  "canAmend": "boolean",
  "canCancel": "boolean",
  "lastAmendmentDate": "datetime (ISO 8601, nullable)",
  "lastCancellationAttempt": "datetime (ISO 8601, nullable)",
  "isAmended": "boolean (default: false)",
  "isCancelled": "boolean (default: false)",
  "originalCertificateId": "string (UUID, nullable)",
  "trader": {
    "traderId": "string (UUID)",
    "traderName": "string (required, max 200)",
    "traderEmail": "string (required, valid email)",
    "traderPhone": "string (required)",
    "traderAddress": {
      "street": "string (required, max 200)",
      "city": "string (required, max 100)",
      "country": "string (required, ISO country code)",
      "postalCode": "string (optional, max 20)"
    }
  },
  "customsBroker": {
    "brokerId": "string (UUID)",
    "brokerName": "string (required, max 200)",
    "brokerEmail": "string (required, valid email)",
    "brokerLicenseNumber": "string (required, max 50)",
    "brokerCompany": "string (required, max 200)",
    "isNominated": "boolean (default: false)",
    "nominationDate": "datetime (ISO 8601, nullable)"
  },
  "shipment": {
    "billOfLadingNumber": "string (required, max 50)",
    "dupNumber": "string (required, max 50)",
    "vesselName": "string (required, max 200)",
    "voyageNumber": "string (required, max 50)",
    "portOfLoading": "string (required, port code)",
    "portOfDischarge": "string (required, port code)",
    "estimatedTimeOfArrival": "datetime (ISO 8601)",
    "actualTimeOfArrival": "datetime (ISO 8601, nullable)"
  },
  "goods": [
    {
      "itemNumber": "integer (sequence)",
```

```

    "description": "string (required, max 500)",
    "hsCode": "string (required, HS code format)",
    "quantity": "number (required, positive)",
    "unit": "string (required, standard units)",
    "weight": "number (required, positive, kg)",
    "volume": "number (optional, positive, m³)",
    "value": "number (required, positive, USD)",
    "origin": "string (required, ISO country code)",
    "manufacturer": "string (optional, max 200)"
  }
],
"containers": [
  {
    "containerNumber": "string (required, max 20)",
    "containerType": "string (required, ISO container type)",
    "sealNumbers": ["string (max 50)"],
    "tareWeight": "number (required, positive, kg)",
    "grossWeight": "number (required, positive, kg)",
    "cargoWeight": "number (required, positive, kg)"
  }
],
"payment": {
  "invoiceNumber": "string (auto-generated)",
  "amount": "number (required, positive, USD)",
  "currency": "string (required, ISO currency code)",
  "paymentStatus": "string (enum)",
  "paymentDate": "datetime (ISO 8601, nullable)",
  "paymentMethod": "string (enum)",
  "refundAmount": "number (optional, positive, USD)",
  "refundStatus": "string (enum, nullable)",
  "refundDate": "datetime (ISO 8601, nullable)"
},
"audit": {
  "createdBy": "string (UUID)",
  "createdDate": "datetime (ISO 8601)",
  "lastModifiedBy": "string (UUID)",
  "lastModifiedDate": "datetime (ISO 8601)",
  "version": "integer (auto-increment)"
}
}

```

Key Enhancements:

- Added `canAmend` and `canCancel` eligibility flags
- Enhanced payment model with refund capabilities
- Added amendment and cancellation tracking fields
- Comprehensive audit trail with versioning

4.2 Amendment Entity Structure

The amendment entity captures all modification requests with complete change tracking:

```

{
  "amendmentId": "string (UUID)",

```

```

"originalCertificateId": "string (UUID, foreign key)",
"amendmentNumber": "string (auto-generated, sequential)",
"requestDate": "datetime (ISO 8601)",
"submittedBy": "string (UUID, foreign key)",
"submittedByRole": "string (enum: CustomsBroker, Trader)",
"status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
"requestReason": "string (required, max 500)",
"businessJustification": "string (optional, max 1000)",
"requestedChanges": [
  {
    "fieldPath": "string (required, JSON path notation)",
    "fieldName": "string (required, human-readable)",
    "oldValue": "any (original value)",
    "newValue": "any (proposed new value)",
    "changeReason": "string (required, max 200)",
    "validationStatus": "string (enum: Valid, Invalid, Warning)"
  }
],
"arcclaReview": {
  "assignedTo": "string (UUID, nullable)",
  "assignedDate": "datetime (ISO 8601, nullable)",
  "reviewStarted": "datetime (ISO 8601, nullable)",
  "reviewCompleted": "datetime (ISO 8601, nullable)",
  "decision": "string (enum: Approved, Rejected, nullable)",
  "decisionReason": "string (optional, max 1000)",
  "reviewComments": "string (optional, max 2000)",
  "impactAssessment": {
    "riskLevel": "string (enum: Low, Medium, High)",
    "affectedSystems": ["string"],
    "complianceImpact": "string (optional, max 500)",
    "financialImpact": "number (optional, USD)"
  }
},
"fees": {
  "amendmentFee": "number (required, positive, USD)",
  "currency": "string (required, ISO currency code)",
  "calculationMethod": "string (required)",
  "invoiceNumber": "string (auto-generated, nullable)",
  "paymentStatus": "string (enum)",
  "paymentDate": "datetime (ISO 8601, nullable)"
},
"processing": {
  "applicationStarted": "datetime (ISO 8601, nullable)",
  "applicationCompleted": "datetime (ISO 8601, nullable)",
  "certificateUpdated": "datetime (ISO 8601, nullable)",
  "notificationsSent": "datetime (ISO 8601, nullable)",
  "processingErrors": [
    {
      "errorCode": "string",
      "errorMessage": "string",
      "errorDate": "datetime (ISO 8601)"
    }
  ]
},
"audit": {
  "createdBy": "string (UUID)",

```

```

    "createdDate": "datetime (ISO 8601)",
    "lastModifiedBy": "string (UUID)",
    "lastModifiedDate": "datetime (ISO 8601)",
    "workflowHistory": [
      {
        "action": "string",
        "performedBy": "string (UUID)",
        "performedDate": "datetime (ISO 8601)",
        "comments": "string (optional)",
        "systemGenerated": "boolean"
      }
    ]
  }
}

```

4.3 Cancellation Entity Structure

The cancellation entity manages certificate termination requests with comprehensive financial tracking:

```

{
  "cancellationId": "string (UUID)",
  "originalCertificateId": "string (UUID, foreign key)",
  "cancellationNumber": "string (auto-generated, sequential)",
  "requestDate": "datetime (ISO 8601)",
  "submittedBy": "string (UUID, foreign key)",
  "submittedByRole": "string (enum: CustomsBroker, Trader)",
  "status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
  "cancellationReason": {
    "reasonCode": "string (enum: required)",
    "reasonDescription": "string (required, max 500)",
    "additionalDetails": "string (optional, max 1000)",
    "urgencyLevel": "string (enum: Normal, High, Critical)"
  },
  "financialImpact": {
    "originalAmount": "number (required, USD)",
    "refundEligibility": "boolean",
    "refundAmount": "number (optional, USD)",
    "cancellationFee": "number (optional, USD)",
    "netRefund": "number (optional, USD)",
    "refundCalculationMethod": "string (required)",
    "refundJustification": "string (optional, max 500)"
  },
  "arcclaReview": {
    "assignedTo": "string (UUID, nullable)",
    "assignedDate": "datetime (ISO 8601, nullable)",
    "reviewStarted": "datetime (ISO 8601, nullable)",
    "reviewCompleted": "datetime (ISO 8601, nullable)",
    "decision": "string (enum: Approved, Rejected, nullable)",
    "decisionReason": "string (optional, max 1000)",
    "reviewComments": "string (optional, max 2000)",
    "riskAssessment": {
      "fraudRisk": "string (enum: Low, Medium, High)",
      "complianceRisk": "string (enum: Low, Medium, High)",
      "operationalRisk": "string (enum: Low, Medium, High)"
    }
  }
}

```

```

    "overallRisk": "string (enum: Low, Medium, High)"
  }
},
"refundProcessing": {
  "refundInitiated": "datetime (ISO 8601, nullable)",
  "refundCompleted": "datetime (ISO 8601, nullable)",
  "refundMethod": "string (enum: nullable)",
  "refundReference": "string (nullable)",
  "refundStatus": "string (enum: nullable)",
  "processingErrors": [
    {
      "errorCode": "string",
      "errorMessage": "string",
      "errorDate": "datetime (ISO 8601)"
    }
  ]
},
"audit": {
  "createdBy": "string (UUID)",
  "createdDate": "datetime (ISO 8601)",
  "lastModifiedBy": "string (UUID)",
  "lastModifiedDate": "datetime (ISO 8601)",
  "workflowHistory": [
    {
      "action": "string",
      "performedBy": "string (UUID)",
      "performedDate": "datetime (ISO 8601)",
      "comments": "string (optional)",
      "systemGenerated": "boolean"
    }
  ]
}
}

```

4.4 Enhanced Status Management

The enhanced status management system provides comprehensive lifecycle tracking with dynamic eligibility determination:

Certificate Status Enumeration:

```

{
  "certificateStatuses": [
    "Draft",
    "Submitted",
    "UnderReview",
    "RequiresAmendment",
    "Approved",
    "PaymentPending",
    "PaymentReceived",
    "Issued",
    "Active",
    "Amended",
    "Cancelled",
    "Expired",
  ]
}

```

```

    "Rejected"
  ]
}

```

Amendment Status Enumeration:

```

{
  "amendmentStatuses": [
    "Pending",
    "UnderReview",
    "RequiresInformation",
    "Approved",
    "Rejected",
    "PaymentPending",
    "PaymentReceived",
    "Processing",
    "Completed",
    "Cancelled",
    "Expired"
  ]
}

```

Cancellation Status Enumeration:

```

{
  "cancellationStatuses": [
    "Pending",
    "UnderReview",
    "RequiresInformation",
    "Approved",
    "Rejected",
    "RefundPending",
    "RefundProcessing",
    "RefundCompleted",
    "Completed",
    "Cancelled"
  ]
}

```

Eligibility Rules Matrix: The system dynamically calculates eligibility flags based on the following business rules:

Amendment Eligibility (`canAmend`):

- Certificate status is in: ["Submitted", "Approved", "Issued", "Active"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was issued within the allowed amendment timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Cancellation Eligibility (`canCancel`):

- Certificate status is in: ["Submitted", "Approved", "PaymentPending", "PaymentReceived", "Issued"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was submitted within the allowed cancellation timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Dynamic Status Calculation: The system recalculates eligibility flags whenever:

- Certificate status changes
 - Amendment or cancellation requests are submitted
 - Time-based rules expire
 - User permissions are modified
 - Business rules are updated by administrators
-

5. Core API Specifications

5.1 Master Data APIs

Master data APIs provide essential reference data required for certificate creation, validation, and business operations.

5.1.1 Cargo Types API

Business Purpose: Manages cargo classification data essential for determining certificate fees, validation rules, and compliance requirements. Critical for proper cargo categorization in international shipping, affecting customs documentation, handling procedures, insurance requirements, and regulatory compliance.

Endpoint: `GET /api/CargoTypes`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	CargoType_Desc
2	CargoType_Desc	String - 200	O	Filter by cargo type description	CONTAINER
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CARGO_UI_001:** CargoType_Desc filter optional, 1-200 characters
- **CARGO_UI_002:** Sort parameter from predefined fields only
- **CARGO_UI_003:** Max 100 records per page

Business Validation Rules:

- **CARGO_BV_001:** Only active cargo types can be used for new CTN creation
- **CARGO_BV_002:** Container cargo requires container details
- **CARGO_BV_003:** Dangerous goods cargo requires special documentation

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique cargo type identifier	1
2	CargoType_Code	String - 10	M	Cargo type code	CONT
3	CargoType_Desc	String - 200	M	Cargo type description	CONTAINER
4	Active	Boolean	M	Active status indicator	true
5	CreatedOn	DateTime	O	Record creation timestamp	2023-01-15T10:30:00Z

Error Codes:

- **CARGO_E001:** "Invalid parameters" (HTTP 400)

- **CARGO_E002:** "No data found" (HTTP 404)
- **CARGO_E003:** "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/CargoTypes?$sort=CargoType_Desc&CargoType_Desc=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "MultiLingualDescription": {
      "Id": 2183,
      "Translations": [
        {
          "Id": 2187,
          "LanguageISO": "en",
          "MultiLingualTextId": 2183,
          "Text": "CONTAINER"
        },
        {
          "Id": 2188,
          "LanguageISO": "fr",
          "MultiLingualTextId": 2183,
          "Text": "CONTENEUR"
        },
        {
          "Id": 2304,
          "LanguageISO": "pt",
          "MultiLingualTextId": 2183,
          "Text": "Contentor"
        }
      ]
    },
    "Code": "CONTAINER",
    "CargoType_Desc": "CONTAINER",
    "CreatedOn": null,
    "CreatedById": null,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "MultiLingualDescription": {
      "Id": 2184,
      "Translations": [
        {
          "Id": 2189,
          "LanguageISO": "en",
          "Text": "BULK"
        }
      ]
    }
  }
]
```

```

    {
      "Id": 2190,
      "LanguageISO": "fr",
      "Text": "VRAC"
    },
    {
      "Id": 2309,
      "LanguageISO": "pt",
      "Text": "Sacrias ou carga fracionada"
    }
  ]
},
"Code": "BULK",
"CargoType_Desc": "BULK",
"CreatedOn": null,
"CreatedById": null,
"ModifiedOn": null,
"ModifiedById": null
}
]

```

Sample JSON Response:

```

[
  {
    "Id": 1,
    "CargoType_Code": "CONT",
    "CargoType_Desc": "CONTAINER",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "CargoType_Code": "BULK",
    "CargoType_Desc": "BULK CARGO",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.2 Incoterms API

Business Purpose: Provides International Commercial Terms (Incoterms) data crucial for determining responsibility boundaries, costs, and risks between buyers and sellers in international trade transactions. Affects shipping responsibilities, insurance requirements, customs clearance obligations, and liability allocation throughout the shipping process.

Endpoint: GET /api/Incoterms

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	IncotermCode
2	IncotermCode	String - 10	O	Filter by Incoterm code	CIF
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **INCO_UI_001:** IncotermCode filter optional, exactly 3 uppercase letters
- **INCO_UI_002:** Display code and description together
- **INCO_UI_003:** Default to "Active Only" for new certificates

Business Validation Rules:

- **INCO_BV_001:** CIF/CIP terms require insurance documentation
- **INCO_BV_002:** Some Incoterms not applicable for certain cargo types
- **INCO_BV_003:** Incoterm selection affects customs valuation

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique Incoterm identifier	2
2	IncotermCode	String - 10	M	Incoterm code	CIF
3	IncotermDesc	String - 200	M	Incoterm description	Cost, Insurance and Freight
4	Active	Boolean	M	Active status indicator	true

Error Codes:

- **INCO_E001:** "Invalid format" (HTTP 400)
- **INCO_E002:** "Not applicable for cargo type" (HTTP 422)
- **INCO_E003:** "Inactive Incoterm" (HTTP 422)

Sample JSON Request:

```
GET /api/Incoterms?$sort=IncotermCode&IncotermCode=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 6,
    "IncotermCode": "CFR",
```

```

    "MultiLingualDescription": {
      "Id": 2211,
      "Translations": [
        {
          "Id": 2238,
          "LanguageISO": "en",
          "MultiLingualTextId": 2211,
          "Text": "Cost and Freight"
        }
      ],
      "DescriptiveFields": "Cost and Freight"
    },
    "Active": true,
    "CreatedOn": null,
    "ModifiedOn": "2023-01-11T16:51:28.8309058",
    "ModifiedById": 1,
    "ModifiedByLogin": "TCNT"
  },
  {
    "Id": 2,
    "IncotermCode": "CIF",
    "MultiLingualDescription": {
      "Id": 2207,
      "Translations": [
        {
          "Id": 2234,
          "LanguageISO": "en",
          "Text": "Cost, Insurance, Freight"
        }
      ],
      "DescriptiveFields": "Cost, Insurance, Freight"
    },
    "Active": true,
    "CreatedOn": null,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.3 Countries API

Business Purpose: Essential for international trade compliance, determining applicable regulations, duties, restrictions, and routing requirements for cargo shipments. Critical for sanctions screening, trade agreement benefits, customs procedures, and ensuring compliance with import/export regulations between trading nations.

Endpoint: `GET /api/Countries`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Country_Name
2	Country_Name	String - 100	O	Filter by country name	Angola

3	ExportingCountry	Boolean	O	Filter exporting countries	true
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **COUNTRY_UI_001:** Country name filter optional, 1-100 characters
- **COUNTRY_UI_002:** Auto-complete dropdown with country name and code
- **COUNTRY_UI_003:** Visual indicator for sanctioned/restricted countries

Business Validation Rules:

- **COUNTRY_BV_001:** Angola must be origin or destination for CNCA certificates
- **COUNTRY_BV_002:** Sanctioned countries require special authorization
- **COUNTRY_BV_003:** Country code must match ISO 3166-1 standard

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique country identifier	10
2	Country_Name	String - 100	M	Country name	Angola
3	Country_Code	String - 3	M	ISO country code	AO
4	ExportingCountry	Boolean	M	Exporting country flag	true
5	ImportingCountry	Boolean	M	Importing country flag	false

Error Codes:

- **COUNTRY_E001:** "Invalid format" (HTTP 400)
- **COUNTRY_E002:** "Not approved for trade" (HTTP 422)
- **COUNTRY_E003:** "Sanctioned country" (HTTP 403)

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Country_Name
2	Country_Name	String - 100	O	Filter by country name	Angola
3	ExportingCountry	Boolean	O	Filter exporting countries	true
4	active	Boolean	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique country identifier	10

2	Country_Name	String - 100	M	Country name	Angola
3	Country_Code	String - 3	M	ISO country code	AO
4	ExportingCountry	Boolean	M	Exporting country flag	true
5	ImportingCountry	Boolean	M	Importing country flag	false

5.1.4 Carriers API

Business Purpose: Manages shipping carrier information essential for maritime transport documentation and tracking. Critical for bill of lading validation, carrier liability determination, vessel scheduling coordination, and ensuring proper documentation of transportation responsibility throughout the supply chain.

Endpoint: `GET /api/Carriers`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 200	O	Filter by carrier name	A.C. ORSSLEFF'S
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CARRIER_UI_001:** Name filter optional, 1-200 characters
- **CARRIER_UI_002:** Display carrier name with city/country for clarity
- **CARRIER_UI_003:** Auto-complete dropdown with carrier search

Business Validation Rules:

- **CARRIER_BV_001:** Only active carriers allowed for new shipments
- **CARRIER_BV_002:** Carrier must be authorized for the route
- **CARRIER_BV_003:** Carrier contact information required for tracking

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique carrier identifier	789
2	Name	String - 200	M	Carrier company name	A.C. ORSSLEFF'S EFTF A/S
3	LicenseNumber	String - 50	O	Carrier license number	LICENSE123
4	Address	String - 500	O	Carrier address	Kongevejen 40
5	City	String - 100	O	Carrier city	2840 Holte
6	CountryId	Integer	O	Country identifier	56

7	Email	String - 200	O	Contact email	chartering@acoe.dk
8	Telephone	String - 50	O	Contact telephone	+45 45 46 00 66
9	TrackingURL	String - 500	O	Tracking website URL	https://track.carrier.com

Error Codes:

- **CARRIER_E001:** "Invalid parameters" (HTTP 400)
- **CARRIER_E002:** "No data found" (HTTP 404)
- **CARRIER_E003:** "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/Carriers?$sort=Name&Name=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1343,
    "LicenseNumber": null,
    "Address": "Care of ZEABORN Ship Management GmbH & Cie KG , Ludwig-Erhard-Strasse 22, 20459 Hamburg, Germany",
    "City": "Hamburg",
    "CountryId": 54,
    "Email": null,
    "Name": "\"E.R. BRISTOL\" Schiffsbeteiligungsgesellschaft",
    "Telephone": null,
    "TrackingURL": null,
    "CreatedOn": "2020-07-17T08:57:52.973",
    "CreatedBy": 1,
    "ModifiedOn": "2020-07-17T08:57:52.973",
    "ModifiedBy": 1
  },
  {
    "Id": 789,
    "LicenseNumber": null,
    "Address": "Kongevej 40",
    "City": "2840 Holte",
    "CountryId": 56,
    "Email": "chartering@acoe.dk",
    "Name": "A.C. ORSSLEFF'S EFTF A/S",
    "Telephone": "+45 45 46 00 66",
    "TrackingURL": "",
    "CreatedOn": "2008-05-13T08:36:14",
    "CreatedBy": 1,
    "ModifiedOn": "2018-08-29T09:24:35.927",
    "ModifiedBy": 1
  }
]
```

```
}  
]
```

5.1.5 Currencies API

Endpoint: `GET /api/Currencies`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Code
2	Code	String - 3	O	Filter by currency code	USD
3	active	Boolean	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique currency identifier	2
2	Code	String - 3	M	ISO currency code	USD
3	Name	String - 100	O	Currency name	US Dollar
4	Symbol	String - 5	O	Currency symbol	\$

Sample JSON Request:

```
GET /api/Currencies?$sort=Code&Code=USD&active=1  
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...  
Accept: application/json
```

Sample JSON Response:

```
[  
  {  
    "Id": 1,  
    "Code": "EUR",  
    "MarkedForUpdate": false,  
    "Countries": [],  
    "Exchange_rates": [],  
    "CreatedOn": null,  
    "CreatedBy": null,  
    "ModifiedOn": null,  
    "ModifiedBy": null  
  },  
  {
```

```

    "Id": 2,
    "Code": "USD",
    "MarkedForUpdate": false,
    "Countries": [],
    "Exchange_rates": [],
    "CreatedOn": null,
    "CreatedById": null,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.6 Banks API

Business Purpose: Manages banking institution data for payment processing and invoice generation in CNCA certificate transactions. Essential for financial settlements, payment validations, and generating proper invoicing documentation with correct banking details.

Endpoint: `GET /api/Banks`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 200	O	Filter by bank name	Standard Bank
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **BANK_UI_001:** Bank name filter optional, 1-200 characters
- **BANK_UI_002:** Display active banks only in dropdown selection
- **BANK_UI_003:** Validate bank selection for payment processing

Business Validation Rules:

- **BANK_BV_001:** Only active banks allowed for new payments
- **BANK_BV_002:** Bank must support international transfers for foreign traders
- **BANK_BV_003:** Bank details required for invoice generation

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique bank identifier	15
2	Name	String - 200	M	Bank institution name	Standard Bank Angola
3	Code	String - 20	O	Bank code	SBA
4	SwiftCode	String - 11	O	International SWIFT code	SBICAOLU

5	Active	Boolean	M	Active status indicator	true
---	--------	---------	---	-------------------------	------

Error Codes:

- **BANK_E001:** "Invalid parameters" (HTTP 400)
- **BANK_E002:** "No data found" (HTTP 404)
- **BANK_E003:** "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/Banks?$sort=Name&Name=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Name": "Banco Angolano de Investimentos",
    "Code": "BAI",
    "Address": "Rua Major Kanhangulo, n° 34",
    "City": "Luanda",
    "CountryID": 10,
    "Country": {
      "Id": 10,
      "Country_Code": "AO",
      "Country_Name": "Angola",
      "ExportingCountry": true,
      "ImportingCountry": true
    },
    "Website": "http://www.bancobai.ao",
    "Telephone": "222 693 800 / 222 693 899",
    "NIFNumber": null,
    "CreatedOn": null,
    "ModifiedOn": "2023-01-24T11:54:44.7181885",
    "ModifiedById": 1,
    "ModifiedByLogin": "TCNT"
  },
  {
    "Id": 2,
    "Name": "Banco Angolano de Negocios e Comercio, S.A.",
    "Code": "BANC",
    "Address": "Travessa da Sorte, n° 12. Maianga",
    "City": "Luanda",
    "CountryID": 10,
    "Country": {
      "Id": 10,
      "Country_Code": "AO",
      "Country_Name": "Angola",
      "ExportingCountry": true,
      "ImportingCountry": true
    }
  }
]
```

```
"Website": null,  
"Telephone": null,  
"NIFNumber": null,  
"CreatedOn": null,  
"ModifiedOn": null,  
"ModifiedById": null  
}  
]
```

- **BANK_E002:** "Bank not authorized for transactions" (HTTP 422)
- **BANK_E003:** "Inactive bank selected" (HTTP 422)

5.1.7 Units API

Business Purpose: Manages measurement unit definitions essential for accurate quantity and weight calculations in cargo documentation. Critical for ensuring consistent unit conversions, compliance with international standards, and accurate cost calculations based on cargo measurements.

Endpoint: `GET /api/Units`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Unit_Desc
2	Unit_Desc	String - 50	O	Filter by unit description	KG
3	UnitType	String - 20	O	Filter by unit category	Weight
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **UNIT_UI_001:** Unit description required, 1-50 characters
- **UNIT_UI_002:** Unit type selection from predefined categories
- **UNIT_UI_003:** Display with conversion factors for clarity

Business Validation Rules:

- **UNIT_BV_001:** Unit must be internationally recognized standard
- **UNIT_BV_002:** Conversion factors must be mathematically accurate
- **UNIT_BV_003:** Weight units required for cargo calculations

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique unit identifier	1
2	Unit_Desc	String - 50	M	Unit description	Kilogram
3	UnitSymbol	String - 10	M	Standard unit symbol	KG

4	UnitType	String - 20	M	Unit category	Weight
5	ConversionFactor	Decimal	O	Conversion to base unit	1.000
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **UNIT_E001:** "Invalid unit parameters" (HTTP 400)
- **UNIT_E002:** "Unit not found" (HTTP 404)
- **UNIT_E003:** "Unit conversion error" (HTTP 422)

Sample JSON Request:

```
GET /api/Units?$sort=Unit_Desc&UnitType=Weight&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Unit_Desc": "Kilogram",
    "UnitSymbol": "KG",
    "UnitType": "Weight",
    "ConversionFactor": 1.000,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Unit_Desc": "Ton",
    "UnitSymbol": "TON",
    "UnitType": "Weight",
    "ConversionFactor": 1000.000,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.8 Container Types API

Business Purpose: Manages container type classifications essential for proper container booking, handling procedures, and shipping cost calculations. Critical for determining container capacity, weight limits, handling requirements, and ensuring accurate freight calculations based on container specifications.

Endpoint: GET /api/ContainerTypes

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Container_Type
2	Container_Type	String - 50	O	Filter by container type	20GP
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CONTAINER_UI_001:** Container type filter optional, 1-50 characters
- **CONTAINER_UI_002:** Display container type with dimensions for clarity
- **CONTAINER_UI_003:** Group standard vs special container types

Business Validation Rules:

- **CONTAINER_BV_001:** Container type must match cargo requirements
- **CONTAINER_BV_002:** Dangerous goods require specialized containers
- **CONTAINER_BV_003:** Weight limits vary by container type

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique container type identifier	1
2	Container_Type	String - 50	M	Container type code	20GP
3	Description	String - 200	M	Container description	20-foot General Purpose
4	Length	Decimal	O	Container length in feet	20.00
5	Width	Decimal	O	Container width in feet	8.00
6	Height	Decimal	O	Container height in feet	8.50
7	MaxWeight	Decimal	O	Maximum weight capacity in tons	28.20
8	Active	Boolean	M	Active status indicator	true

Error Codes:

- **CONTAINER_E001:** "Invalid parameters" (HTTP 400)
- **CONTAINER_E002:** "Container type not found" (HTTP 404)
- **CONTAINER_E003:** "Container type not applicable for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/ContainerTypes?$sort=Container_Type&Container_Type=20GP&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Container_Type": "20GP",
    "Description": "20-foot General Purpose",
    "Length": 20.00,
    "Width": 8.00,
    "Height": 8.50,
    "MaxWeight": 28.20,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Container_Type": "40GP",
    "Description": "40-foot General Purpose",
    "Length": 40.00,
    "Width": 8.00,
    "Height": 8.50,
    "MaxWeight": 30.48,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.9 Transport Types API

Business Purpose: Manages transportation mode classifications essential for logistics planning, route optimization, and freight calculations. Critical for determining shipping costs, transit times, regulatory requirements, and ensuring proper transport documentation based on cargo and route characteristics.

Endpoint: GET /api/TransportTypes

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Type_Name
2	Type_Name	String - 50	O	Filter by transport type	Sea

3	active	Boolean	O	Filter active records only	1
---	--------	---------	---	----------------------------	---

UI Validation Rules:

- **TRANSPORT_UI_001:** Transport type filter optional, 1-50 characters
- **TRANSPORT_UI_002:** Display with appropriate transport icons
- **TRANSPORT_UI_003:** Group by transport mode for clarity

Business Validation Rules:

- **TRANSPORT_BV_001:** Transport type must match cargo requirements
- **TRANSPORT_BV_002:** Dangerous goods require specific transport modes
- **TRANSPORT_BV_003:** Some goods restricted for certain transport types

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique transport type identifier	1
2	Type_Name	String - 50	M	Transport type name	Sea
3	Description	String - 200	O	Transport type description	Maritime sea transport
4	Code	String - 10	O	Transport type code	SEA
5	Active	Boolean	M	Active status indicator	true

Error Codes:

- **TRANSPORT_E001:** "Invalid parameters" (HTTP 400)
- **TRANSPORT_E002:** "Transport type not found" (HTTP 404)
- **TRANSPORT_E003:** "Transport type not applicable for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/TransportTypes?$sort=Type_Name&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Type_Name": "Air",
    "Description": "Air cargo transport",
    "Code": "AIR",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
  }
]
```

```

    "ModifiedById": null
  },
  {
    "Id": 2,
    "Type_Name": "Sea",
    "Description": "Maritime sea transport",
    "Code": "SEA",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.10 Locations/Ports API

Business Purpose: Manages geographical location and port information essential for origin/destination tracking, routing decisions, and customs compliance. Critical for calculating shipping routes, determining applicable regulations, port charges, and ensuring accurate geographic references in certificates and shipping documents.

Endpoint: `GET /api/Locations`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	PortName
2	CountryId	Integer	M	Country identifier filter	4
3	PortName	String - 200	O	Filter by port name	Dubai
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **LOCATION_UI_001:** Country ID required for location filtering
- **LOCATION_UI_002:** Port name search minimum 2 characters
- **LOCATION_UI_003:** Display with country name for clarity

Business Validation Rules:

- **LOCATION_BV_001:** Location must be active port for shipping
- **LOCATION_BV_002:** Port must support container operations
- **LOCATION_BV_003:** Location must have valid customs facilities

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique location identifier	1
2	PortName	String - 200	M	Port/location name	Port of Dubai

3	PortCode	String - 10	O	Standard port code	DXB
4	CountryId	Integer	M	Associated country	4
5	CountryName	String - 100	O	Country name	United Arab Emirates
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **LOCATION_E001:** "Invalid country ID" (HTTP 400)
- **LOCATION_E002:** "Port not found" (HTTP 404)
- **LOCATION_E003:** "Port not operational for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/Locations?$sort=PortName&CountryId=4&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "PortName": "Port of Dubai",
    "PortCode": "DXB",
    "CountryId": 4,
    "CountryName": "United Arab Emirates",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "PortName": "Abu Dhabi Port",
    "PortCode": "AUH",
    "CountryId": 4,
    "CountryName": "United Arab Emirates",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.11 Goods Classifications API

Business Purpose: Manages standardized goods classification codes essential for customs clearance, duty calculations, and regulatory compliance. Critical for determining applicable tariffs, import restrictions, statistical reporting, and ensuring proper cargo categorization according to international trade standards.

Endpoint: `GET /api/GoodsClassifications`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Goods_Classification_Desc
2	Goods_Classification_Desc	String - 200	O	Filter by classification	Electronics
3	ClassificationCode	String - 20	O	Filter by classification code	HS8517
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **GOODS_UI_001:** Classification filter minimum 3 characters
- **GOODS_UI_002:** Display with HS code for clarity
- **GOODS_UI_003:** Group by classification category

Business Validation Rules:

- **GOODS_BV_001:** Classification must be valid HS code
- **GOODS_BV_002:** Some goods require special permits
- **GOODS_BV_003:** Prohibited goods cannot be classified

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique classification identifier	1
2	Goods_Classification_Desc	String - 200	M	Classification description	Electronic equipment
3	ClassificationCode	String - 20	M	HS classification code	HS8517
4	Category	String - 100	O	Classification category	Electronics
5	DutyRate	Decimal	O	Applicable duty rate percentage	5.00
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **GOODS_E001:** "Invalid classification parameters" (HTTP 400)
- **GOODS_E002:** "Classification not found" (HTTP 404)
- **GOODS_E003:** "Classification not applicable" (HTTP 422)

Sample JSON Request:

```
GET /api/GoodsClassifications?$sort=Goods_Classification_Desc&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Goods_Classification_Desc": "Electronic equipment",
    "ClassificationCode": "HS8517",
    "Category": "Electronics",
    "DutyRate": 5.00,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Goods_Classification_Desc": "Textile products",
    "ClassificationCode": "HS6203",
    "Category": "Textiles",
    "DutyRate": 12.00,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.12 IMO Codes API

Business Purpose: Manages International Maritime Organization dangerous goods classification codes essential for hazardous cargo handling, safety compliance, and regulatory adherence. Critical for determining special handling requirements, storage restrictions, transport documentation, and ensuring maritime safety standards compliance.

Endpoint: `GET /api/IMOs`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	IMO_Desc

2	IMO_Desc	String - 200	O	Filter by IMO description	Flammable Liquids
3	IMOCClass	String - 10	O	Filter by IMO class	3
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **IMO_UI_001:** IMO description filter minimum 3 characters
- **IMO_UI_002:** Display with hazard class for safety
- **IMO_UI_003:** Highlight dangerous goods with warning indicators

Business Validation Rules:

- **IMO_BV_001:** IMO class must be valid IMDG code
- **IMO_BV_002:** Dangerous goods require special documentation
- **IMO_BV_003:** Some IMO classes prohibited for certain routes

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique IMO identifier	1
2	IMO_Desc	String - 200	M	IMO description	Flammable Liquids
3	IMOCClass	String - 10	M	IMDG hazard class	3
4	UNNumber	String - 10	O	UN identification number	UN1203
5	PackingGroup	String - 5	O	Packing group classification	II
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **IMO_E001:** "Invalid IMO parameters" (HTTP 400)
- **IMO_E002:** "IMO code not found" (HTTP 404)
- **IMO_E003:** "Dangerous goods not permitted" (HTTP 422)

Sample JSON Request:

```
GET /api/IMOs?$sort=IMO_Desc&IMOCClass=3&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "IMO_Desc": "Flammable Liquids",
```

```
    "IMOCClass": "3",
    "UNNumber": "UN1203",
    "PackingGroup": "II",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "IMO_Desc": "Corrosive Substances",
    "IMOCClass": "8",
    "UNNumber": "UN1760",
    "PackingGroup": "II",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.13 Vessels API

Business Purpose: Manages vessel information essential for maritime transport operations, vessel tracking, and shipping documentation. Critical for identifying vessels in certificates, tracking cargo movements, ensuring vessel compliance, and providing accurate shipping details for customs and logistics operations.

Endpoint: `GET /api/Vessels`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 256	O	Filter by vessel name	CPO HAMBURG
3	Code	String - 20	O	Filter by vessel code	229638000
4	active	Integer	O	Active status filter	1

UI Validation Rules:

- **VESSEL_UI_001:** Vessel name filter minimum 3 characters
- **VESSEL_UI_002:** Display with IMO number for identification
- **VESSEL_UI_003:** Show flag state for vessel information

Business Validation Rules:

- **VESSEL_BV_001:** Vessel must have valid IMO number
- **VESSEL_BV_002:** Vessel must be registered and active
- **VESSEL_BV_003:** Vessel must be suitable for cargo type

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique vessel identifier	5132
2	Name	String - 256	M	Vessel name	CPO HAMBURG1234
3	Code	String - 20	M	Vessel identification code	229638000
4	IDNumber	String - 20	M	IMO number	9450375
5	FlagId	Integer	O	Flag state identifier	148
6	Flag	String - 100	O	Flag state name	Germany
7	Active	Boolean	M	Active status	true

Error Codes:

- **VESSEL_E001:** "Invalid vessel parameters" (HTTP 400)
- **VESSEL_E002:** "Vessel not found" (HTTP 404)
- **VESSEL_E003:** "Vessel not available for route" (HTTP 422)

Sample JSON Request:

```
GET /api/Vessels?$sort=Name&Name=CPO&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 5132,
    "Name": "CPO HAMBURG1234",
    "Code": "229638000",
    "IDNumber": "9450375",
    "FlagId": 148,
    "Flag": "Germany",
    "Active": true,
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": "2023-01-25T13:18:15.1194343",
    "ModifiedById": 13275,
    "ModifiedByLogin": "TESTEXCONSUL"
  },
  {
    "Id": 5163,
    "Name": "NILEDUTCH LION",
    "Code": "636013690",
```

```

    "IDNumber": "9337456",
    "FlagId": 128,
    "Flag": "Liberia",
    "Active": true,
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  }
}

```

5.1.14 CTN Cities API

Business Purpose: Manages city information essential for origin and destination tracking in certificate processing. Critical for accurate geographical references, determining applicable regulations, calculating logistics routes, and ensuring precise location data for customs compliance and certificate validation.

Endpoint: `GET /api/CTNCities`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	CountryId	Integer	M	Country identifier filter	10
3	Name	String - 200	O	Filter by city name	Luanda
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CITY_UI_001:** Country ID required for city filtering
- **CITY_UI_002:** City name search minimum 2 characters
- **CITY_UI_003:** Display with country name for clarity

Business Validation Rules:

- **CITY_BV_001:** City must exist within selected country
- **CITY_BV_002:** City must be active for trade operations
- **CITY_BV_003:** City must support logistics operations

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique city identifier	262
2	Name	String - 200	M	City name	Benguela

3	CountryId	Integer	M	Associated country identifier	10
4	Country	String - 100	O	Country name	Angola
5	Active	Boolean	M	Active status indicator	true

Error Codes:

- **CITY_E001:** "Invalid country ID" (HTTP 400)
- **CITY_E002:** "City not found" (HTTP 404)
- **CITY_E003:** "City not operational for trade" (HTTP 422)

Sample JSON Request:

```
GET /api/CTNCities?$sort=Name&CountryId=10&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 262,
    "Active": true,
    "CountryId": 10,
    "Name": "Benguela",
    "Country": "Angola",
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  },
  {
    "Id": 277,
    "Active": true,
    "CountryId": 10,
    "Name": "Caála",
    "Country": "Angola",
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  },
  {
    "Id": 255,
    "Active": true,
    "CountryId": 10,
    "Name": "Cabinda",
    "Country": "Angola",
```

```

    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  }
]

```

5.1.15 CTN Ports API

Business Purpose: Manages Angolan port information essential for CTN certificate processing and customs clearance. Critical for determining applicable port authorities, customs offices, handling procedures, and ensuring accurate port references in certificates and logistics documentation.

Endpoint: `GET /api/CTNPorts`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 100	O	Filter by port name	Luanda
3	PortCode	String - 10	O	Filter by port code	LAD
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CTNPORT_UI_001:** Port name search minimum 2 characters
- **CTNPORT_UI_002:** Display with customs office information
- **CTNPORT_UI_003:** Show operational status clearly

Business Validation Rules:

- **CTNPORT_BV_001:** Port must be operational for CTN processing
- **CTNPORT_BV_002:** Customs office must be active
- **CTNPORT_BV_003:** Port must support container operations

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique port identifier	1
2	Name	String - 100	M	Port name	Port of Luanda
3	PortCode	String - 10	M	Standard port code	LAD
4	CustomsOffice	String - 100	O	Associated customs office	Alfândega de Luanda

5	Active	Boolean	M	Active status indicator	true
---	--------	---------	---	-------------------------	------

Error Codes:

- **CTNPORT_E001:** "Invalid port parameters" (HTTP 400)
- **CTNPORT_E002:** "Port not found" (HTTP 404)
- **CTNPORT_E003:** "Port not operational" (HTTP 422)

Sample JSON Request:

```
GET /api/CTNPorts?$sort=Name&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Name": "Port of Luanda",
    "PortCode": "LAD",
    "CustomsOffice": "Alfândega de Luanda",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Name": "Port of Lobito",
    "PortCode": "LBT",
    "CustomsOffice": "Alfândega de Lobito",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.2 Certificate Management APIs

Certificate management APIs handle the core CTN certificate lifecycle operations, enabling the creation, retrieval, and management of CNCA certificates throughout their entire lifecycle from draft to completion.

Note: APIs marked as "Internal to JUL" are handled within the JUL system and do not require SINTECE implementation.

5.2.1 CTN List API (Internal to JUL)

Note: This API is internal to the JUL system and does not require implementation by SINTECE.

Business Purpose: Provides comprehensive listing and filtering capabilities for CTN certificates. Essential for dashboard displays, search functionality, status tracking, and bulk operations. Supports real-time monitoring of certificate processing status and enables efficient certificate portfolio management for traders and customs brokers.

Endpoint: `GET /api/ctns`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand for related entities	CargoType,Status,Visum_Agent,CreatedBy,Consignee
2	\$sort	String - 100	O	OData sort parameter	-ModifiedOn
3	\$top	Integer	O	Number of records to return	10
4	\$filter	String - 1000	O	OData filter expression	Status eq 'Active'

UI Validation Rules:

- **CTN_UI_001:** \$top parameter max 100 records per request
- **CTN_UI_002:** \$expand parameter validate against allowed entity names
- **CTN_UI_003:** \$filter parameter OData syntax validation

Business Validation Rules:

- **CTN_BV_001:** User can only view certificates belonging to their organization
- **CTN_BV_002:** Sensitive data requires role-based access
- **CTN_BV_003:** Date range filters max 2 years for performance

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique CTN identifier	503808
2	CTN_Reference_Number	String - 50	M	System generated CTN reference	AO-CNT-503808-2023
3	BL_number	String - 50	M	Bill of lading number	vc568009iujh
4	UniqueTradeNumber	String - 50	M	Unique trade number (DUP)	56789098765
5	DCNumber	String - 50	O	Declaration certificate number	7777777
6	StatusId	Integer	M	Certificate status identifier	1

7	CargoTypeid	Integer	M	Cargo type identifier	1
8	Total_Value_Of_Goods	Decimal	M	Total value of goods in USD	15000.00
9	CreatedOn	DateTime	M	Record creation timestamp	2025-11-12T18:49:30.88Z
10	ModifiedOn	DateTime	O	Record modification timestamp	2025-11-12T19:15:45.23Z

Error Codes:

- **CTN_E001:** "Invalid parameters" (HTTP 400)
- **CTN_E002:** "Access denied" (HTTP 403)
- **CTN_E003:** "No data found" (HTTP 404)

Sample JSON Request:

```
GET /api/ctns?$expand=CargoType,Status,Visum_Agent,CreatedBy,Consignee&$sort=-ModifiedOn&$top=10
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 503808,
    "CTN_Reference_Number": "170543",
    "StatusId": 2,
    "Groupage": false,
    "ParentCTNId": null,
    "CargoTypeId": 1,
    "ETD": "2025-11-12T00:00:00",
    "ETA": "2025-12-09T00:00:00",
    "BL_number": "vc568009iujh",
    "IncotermId": 2,
    "OriginCountryId": 10,
    "FinalDestinationCountryId": 4,
    "FreightPaymentTypeId": 1,
    "Total_number_containers": 1,
    "Total_number_vehicles": 0,
    "Total_Ocean_Freight": 22.0,
    "Total_Value_Of_Goods": 33.0,
    "Total_Charges": 0.0,
    "General_Total": 22.0,
    "View_CurrencyId": 1,
    "Exchange_Rate": 1.0,
    "UniqueTradeNumber": "56789098765",
    "DCNumber": "7777777",
    "DateRequestVisa": "2025-11-12T19:41:07.533",
    "IsExport": true,
    "IsImport": false,
  }
]
```

```
"CargoType": {
  "Id": 1,
  "CargoType_Desc": "CONTAINER",
  "Code": "CONTAINER"
},
"Status": {
  "Id": 2,
  "Status_Desc": "Submitted",
  "Code": "SUBMITTED"
},
"OriginCountry": {
  "Id": 10,
  "Country_Name": "Angola",
  "Country_Code": "AO"
},
"FinalDestinationCountry": {
  "Id": 4,
  "Country_Name": "Afghanistan",
  "Country_Code": "AF"
},
"CreatedOn": "2025-11-12T18:49:30.8817831+01:00",
"ModifiedOn": "2025-11-12T19:41:07.533",
"CreatedBy": 1
}
```

5.2.2 CTN Details API (Internal to JUL)

Note: This API is internal to the JUL system and does not require implementation by SINTECE.

Business Purpose: Retrieves comprehensive CTN certificate details including all related entities and status information essential for certificate management and processing. Critical for viewing complete CTN records, managing certificate lifecycle, tracking status changes, and providing detailed information for customs clearance and logistics coordination.

Endpoint: GET /api/ctns/{id}

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN unique identifier from URL path	503808
2	\$expand	String - 500	O	OData expand for related entities	CargoType,ParentCTN,Status,Incoterm

UI Validation Rules:

- **CTNDET_UI_001:** CTN ID must be valid integer for lookup
- **CTNDET_UI_002:** Display expanded entities in structured format
- **CTNDET_UI_003:** Show status progression and timeline

Business Validation Rules:

- **CTNDET_BV_001:** CTN must exist and be accessible to user
- **CTNDET_BV_002:** User must have read permission for CTN
- **CTNDET_BV_003:** Related entities must be properly expanded

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique CTN identifier	503808
2	CTN_Reference_Number	String - 50	M	System generated CTN reference	170543
3	BL_number	String - 50	M	Bill of lading number	vc568009iujh
4	UniqueTradeNumber	String - 50	M	Unique trade number	56789098765
5	VoyageNo	String - 50	O	Voyage number	VOY123
6	StatusId	Integer	M	Current status identifier	2
7	Total_number_containers	Integer	M	Total number of containers	1
8	Total_Value_Of_Goods	Decimal	M	Total value of goods	33.0
9	CTNCost	Decimal	O	Certificate cost	0.0
10	ETD	DateTime	M	Estimated time of departure	2025-11-12T00:00:00
11	ETA	DateTime	M	Estimated time of arrival	2025-12-09T00:00:00

Error Codes:

- **CTNDET_E001:** "CTN not found" (HTTP 404)
- **CTNDET_E002:** "Access denied to CTN" (HTTP 403)
- **CTNDET_E003:** "Invalid expand parameters" (HTTP 400)

Sample JSON Request:

```
GET /api/ctns/503808?
$expand=CargoType,ParentCTN,Status,Incoterm,Carrier,OriginCountry,Origin_City,FinalDestinationCountry,Fr

Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```

{
  "CTN_Addresses": [],
  "CTN_Containers": [],
  "CTN_Goods": [],
  "CTN_RORO": [],
  "CTN_Tracking": [],
  "Id": 503808,
  "CTN_Reference_Number": "170543",
  "StatusId": 2,
  "Groupage": false,
  "ParentCTNId": null,
  "CargoTypeId": 1,
  "ETD": "2025-11-12T00:00:00",
  "ETA": "2025-12-09T00:00:00",
  "BL_number": "vc568009iujh",
  "IncotermId": 2,
  "OriginCountryId": 10,
  "FinalDestinationCountryId": 4,
  "FreightPaymentTypeId": 1,
  "Total_number_containers": 1,
  "Total_number_vehicles": 0,
  "Total_Ocean_Freight": 22.0,
  "Total_Value_Of_Goods": 33.0,
  "Total_Charges": 0.0,
  "General_Total": 22.0,
  "View_CurrencyId": 1,
  "Exchange_Rate": 1.0,
  "UniqueTradeNumber": "56789098765",
  "CTNCost": 0.0,
  "CommissionCNC": 0.00,
  "Status": {
    "Id": 2,
    "MultiLingualDescription": {
      "Id": 2195,
      "Translations": [
        {
          "Id": 2211,
          "LanguageISO": "en",
          "MultiLingualTextId": 2195,
          "Text": "Request Visa"
        }
      ]
    },
    "Code": "RequestVisa"
  },
  "Origin_City": {
    "Id": 262,
    "Active": true,
    "CountryId": 10,
    "Name": "Benguela"
  },
  "RefusedBy": {
    "Id": 13267,
    "Official_Name": "CNC LUANDA EXPORT SUBAGENT",
    "Email": "helpdesk@tcnt.eu",

```

```
"Address": "LUANDA",  
"Active": true  
},  
"CreatedOn": "2025-11-12T18:58:11.807",  
"CreatedById": 13267,  
"ModifiedOn": "2025-11-12T19:32:43.48",  
"ModifiedById": 13267  
}
```

5.2.3 CTN Attachments API

Business Purpose: Manages CTN certificate attachments and supporting documents essential for compliance verification and audit trails. Critical for storing required documentation, enabling file downloads, maintaining document integrity, and ensuring complete certificate packages for customs and regulatory review.

Endpoint: `GET /api/ctnAttachments`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 300	O	OData expand parameter	CreatedBy,attachment,attachmentName
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (alternative)	503808

UI Validation Rules:

- **CTNATT_UI_001:** CTN ID required for attachment lookup
- **CTNATT_UI_002:** Display file types with appropriate icons
- **CTNATT_UI_003:** Show file size and upload date for management

Business Validation Rules:

- **CTNATT_BV_001:** CTN must exist and be accessible
- **CTNATT_BV_002:** File size must not exceed system limits
- **CTNATT_BV_003:** File types must be approved formats

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Attachment record ID	12345
2	CTNId	Integer	M	Related CTN ID	503808
3	FileName	String - 255	M	Original file name	bill_of_lading.pdf
4	FileSize	Integer	M	File size in bytes	2048576

5	AttachmentTypeId	Integer	M	Type of attachment	1
6	AttachmentTypeName	String - 100	O	Attachment type description	Bill of Lading
7	UploadDate	DateTime	M	File upload timestamp	2023-12-15T10:30:00Z
8	CreatedById	Integer	M	User who uploaded file	13345

Error Codes:

- **CTNATT_E001:** "Invalid CTN ID" (HTTP 400)
- **CTNATT_E002:** "No attachments found" (HTTP 404)
- **CTNATT_E003:** "File access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/ctnAttachments?$expand=CreatedBy,attachment,attachmentName&ctnid=503808
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 12345,
    "CTNId": 503808,
    "FileName": "bill_of_lading.pdf",
    "FileSize": 2048576,
    "AttachmentTypeId": 1,
    "AttachmentTypeName": "Bill of Lading",
    "UploadDate": "2023-12-15T10:30:00Z",
    "CreatedById": 13345,
    "CreatedBy": {
      "Id": 13345,
      "UserName": "CNCEXPORTER",
      "Email": "exporter@company.com"
    }
  }
]
```

5.2.4 CTN Creation API (Certificate Submission)

Note: This unified API consolidates what was previously multiple separate operations (creation, addresses, goods, containers, tracking, attachments) into one comprehensive submission. JUL will handle all data processing internally and save as draft, eliminating the need for multiple API calls from SINTECE.

Business Purpose: Creates new CTN certificate records with complete information for CNCA certificate applications in a single comprehensive request. This unified API combines what was previously multiple separate operations (creation, addresses, goods, containers, tracking, attachments) into one streamlined submission. Critical for submitting complete certificate requests, establishing certificate records with all required data, validating comprehensive submission data, and enabling the certificate approval workflow efficiently.

Endpoint: **POST** /api/ctns

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
Core CTN Information					
1	Id	Integer	O	CTN identifier (null for new)	null
2	CTN_Reference_Number	String - 50	O	System generated reference	""
3	StatusId	Integer	M	Status identifier (1=Created)	1
4	Groupage	Boolean	O	Groupage indicator	false
5	ParentCTNId	Integer	O	Parent CTN identifier	null
6	CargoTypeId	Integer	M	Cargo type identifier	1
7	ETD	DateTime	O	Estimated time departure	null
8	ETA	DateTime	O	Estimated time arrival	null
9	BL_number	String - 50	M	Bill of lading number	vc568009iujh
10	IncotermdId	Integer	M	Incoterm identifier	2
11	OriginCountryId	Integer	M	Origin country identifier	10
12	FinalDestinationCountryId	Integer	M	Destination country identifier	4
13	FreightPaymentTypeId	Integer	M	Freight payment type identifier	1
14	Total_number_containers	Integer	O	Total containers count	0
15	Total_number_vehicles	Integer	O	Total vehicles count	0

16	Total_Ocean_Freight	Decimal	O	Total ocean freight value	0
17	Total_Value_Of_Goods	Decimal	O	Total goods value	0
18	Total_Charges	Decimal	O	Total charges amount	0
19	General_Total	Decimal	O	General total amount	0
20	View_CurrencyId	Integer	M	Currency identifier	1
21	Exchange_Rate	Decimal	O	Exchange rate	1
22	Visum_Location	String - 100	O	Visa location	null
23	Visum_Reference_number	String - 50	O	Visa reference number	null
24	Visum_Cost	Decimal	O	Visa cost	1
25	Visum_Date	DateTime	O	Visa date	null
26	Visum_AgentId	Integer	O	Visa agent identifier	null
27	Invoice_to_agent	String - 100	O	Invoicing agent	null
28	VoyageNo	String - 50	O	Voyage number	null
29	RegularisationCTN	Boolean	O	Regularisation flag	false
30	IsExport	Boolean	M	Export indicator	true
31	IsImport	Boolean	O	Import indicator	false
32	Origin_CityId	Integer	M	Origin city identifier	262
33	Final_Destination_CityId	Integer	O	Final destination city	null
34	CarrierId	Integer	M	Carrier identifier	789
35	UniqueTradeNumber	String - 50	M	Unique trade number	56789098765
36	Commission	Decimal	O	Commission amount	0
37	CommissionCNC	Decimal	O	CNC commission	0

38	CTNExchangeRate	Decimal	O	CTN exchange rate	null
39	MarkedForUpdate	Boolean	O	Update marker	false
40	ReceivedByGumar	Boolean	O	Gumar receipt flag	false
41	ErrorSendToGumar	Boolean	O	Gumar error flag	false
42	BankId	Integer	O	Bank identifier	2
43	Consigneeld	Integer	M	Consignee identifier	43214
44	ReExport	Boolean	O	Re-export indicator	false
45	DCNumber	String - 50	O	Document control number	7777777
46	CreatedOn	DateTime	O	Creation timestamp	2025-11-12T18:49:30.88Z
47	CreatedByld	Integer	O	Creator identifier	1
Address Information					
48	CTN_Addresses	Array	O	Array of address objects for shipper, consignee, forwarder, notify party	[...]
48.1	- Id	Integer	O	Address identifier (null for new)	null
48.2	- QryAddressTypeId	Integer	M	Address type (1=Shipper, 2=Consignee, 3=Forwarder, 4=Notify)	2
48.3	- Name	String - 200	M	Company/Person name	M.A.C.S AS.TEC.N. E IND
48.4	- Address	String - 500	M	Physical address	maculusso rua comandate
48.5	- City	String - 100	M	City name	kwenhan
48.6	- CountryId	Integer	M	Country identifier	10

48.7	- Email	String - 100	O	Email address	contact@company.com
48.8	- Telephone	String - 50	M	Phone number	923798669
48.9	- NIFNumber	String - 50	M	Tax identification number	0000000458608
Goods Information					
49	CTN_Goods	Array	O	Array of goods/cargo objects	[...]
49.1	- Id	Integer	O	Goods identifier (null for new)	null
49.2	- GoodsClassificationId	Integer	M	Goods classification identifier	16882
49.3	- IMOld	Integer	M	IMO hazardous code identifier	2
49.4	- CargoTypeId	Integer	M	Cargo type identifier	1
49.5	- DescriptionGoods	String - 500	M	Goods description	test
49.6	- GrossWeightKG	Decimal	M	Gross weight in kg	10.0
49.7	- VolumeCBM	Decimal	M	Volume in cubic meters	11.0
49.8	- OceanFreight	Decimal	M	Ocean freight cost	22.0
49.9	- ValueOfGoods	Decimal	M	Value of goods	33.0
49.10	- NumberOfPackages	Integer	M	Number of packages	3
Container Information					
50	CTN_Containers	Array	O	Array of container objects	[...]
50.1	- Id	Integer	O	Container identifier (null for new)	null
50.2	- ContainerTypeId	Integer	M	Container type identifier	31

50.3	- Number_of_Containers	Integer	M	Number of containers	1
50.4	- ContainerNumbers	String - 100	M	Container number(s)	MSCU1234567
50.5	- SealNumbers	String - 100	M	Seal number(s)	23234
50.6	- Groupage	Boolean	O	Groupage indicator	false
50.7	- IsEmpty	Boolean	O	Empty container indicator	false
50.8	- OwnedByShipper	Boolean	O	Shipper owned indicator	false
Transport/Tracking Information					
51	CTN_Tracking	Array	O	Array of transport route objects	[...]
51.1	- Id	Integer	O	Tracking identifier	null
51.2	- SourceCountryId	Integer	M	Source country identifier	10
51.3	- SourcePortId	Integer	M	Source port identifier	80
51.4	- ETD	DateTime	M	Estimated time departure	2025-11-12T00:00:00.000Z
51.5	- TransportTypeId	Integer	M	Transport type identifier	1
51.6	- VoyageNumber	String - 50	M	Voyage number	546894
51.7	- CarrierId	Integer	M	Carrier identifier	789
51.8	- VesselId	Integer	M	Vessel identifier	5163
51.9	- DestinationCountryId	Integer	M	Destination country identifier	4
51.10	- DestinationPortId	Integer	M	Destination port identifier	35
51.11	- ETA	DateTime	M	Estimated time arrival	2025-12-09T00:00:00.000Z

51.12	- Sequence	Integer	M	Sequence number	1
RORO Information					
52	CTN_RORO	Array	O	Array of RORO cargo objects (for roll-on/roll-off cargo)	[...]
Attachment Information					
53	CTN_Attachments	Array	O	Array of document attachment objects	[...]
53.1	- Id	Integer	O	Attachment identifier (null for new)	null
53.2	- AttachmentGuid	String - 50	M	File upload GUID from fileupload API	0da7461c-2155-4d20-a695-6b7463367327
53.3	- NameId	Integer	M	Attachment name type identifier	4

UI Validation Rules:

- **CTNCRE_UI_001:** All mandatory fields must be completed for core CTN information
- **CTNCRE_UI_002:** BL number must be unique and valid format
- **CTNCRE_UI_003:** Trade number must follow required pattern
- **CTNCRE_UI_004:** At least one address must be provided (typically consignee)
- **CTNCRE_UI_005:** All address fields must be complete when address is provided
- **CTNCRE_UI_006:** At least one goods entry must be provided for the shipment
- **CTNCRE_UI_007:** Container information must match declared container count
- **CTNCRE_UI_008:** Transport route must have valid origin and destination
- **CTNCRE_UI_009:** Attachment GUIDs must be valid from fileupload API

Business Validation Rules:

- **CTNCRE_BV_001:** BL number must not exist in system
- **CTNCRE_BV_002:** All referenced entities (countries, ports, carriers, etc.) must be valid and active
- **CTNCRE_BV_003:** User must have permission to create CTNs
- **CTNCRE_BV_004:** NIFNumber validation must pass for Angola addresses
- **CTNCRE_BV_005:** Total values must match sum of individual goods values
- **CTNCRE_BV_006:** Container count must match actual container entries
- **CTNCRE_BV_007:** Transport sequence must be logical (ETD before ETA)
- **CTNCRE_BV_008:** Required attachments must be present based on trade type

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	New CTN identifier	503808

2	CTN_Reference_Number	String - 50	M	System generated reference	AO-CNT-503808-2025
3	StatusId	Integer	M	Initial status (1=Draft)	1
4	BL_number	String - 50	M	Confirmed BL number	vc568009iujh
5	UniqueTradeNumber	String - 50	M	Confirmed trade number	56789098765
6	CreatedOn	DateTime	M	Creation timestamp	2025-11-14T10:30:00Z
7	CreatedById	Integer	M	Creator user identifier	1
8	AddressesCreated	Integer	M	Number of addresses created	2
9	GoodsCreated	Integer	M	Number of goods entries created	1
10	ContainersCreated	Integer	M	Number of container entries created	1
11	TrackingCreated	Integer	M	Number of tracking entries created	1
12	AttachmentsCreated	Integer	M	Number of attachments created	2

Error Codes:

- **CTNCRE_E001:** "BL number already exists" (HTTP 409)
- **CTNCRE_E002:** "Invalid reference data" (HTTP 400)
- **CTNCRE_E003:** "Insufficient permissions" (HTTP 403)
- **CTNCRE_E004:** "Invalid address information" (HTTP 422)
- **CTNCRE_E005:** "Invalid goods information" (HTTP 422)
- **CTNCRE_E006:** "Invalid container information" (HTTP 422)
- **CTNCRE_E007:** "Invalid tracking information" (HTTP 422)
- **CTNCRE_E008:** "Invalid attachment information" (HTTP 422)
- **CTNCRE_E009:** "NIF validation failed" (HTTP 422)

Sample JSON Request:

```
POST /api/ctns
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

{
  "Id": null,
  "CTN_Reference_Number": "",
  "StatusId": 1,
  "Groupage": false,
  "ParentCTNId": null,
  "CargoTypeId": 1,
```

```
"ETD": null,
"ETA": null,
"BL_number": "vc568009iujh",
"IncotermId": 2,
"OriginCountryId": 10,
"FinalDestinationCountryId": 4,
"FreightPaymentTypeId": 1,
"Total_number_containers": 1,
"Total_number_vehicles": 0,
"Total_Ocean_Freight": 22.0,
"Total_Value_Of_Goods": 33.0,
"Total_Charges": 55.0,
"General_Total": 55.0,
"View_CurrencyId": 1,
"Exchange_Rate": 1,
"Visum_Location": null,
"Visum_Reference_number": null,
"Visum_Cost": 1,
"Visum_Date": null,
"Visum_AgentId": null,
"Invoicer_to_agent": null,
"VoyageNo": "546894",
"RegularisationCTN": false,
"IsExport": true,
"IsImport": false,
"Origin_CityId": 262,
"Final_Destination_CityId": null,
"CarrierId": 789,
"UniqueTradeNumber": "56789098765",
"Commission": 0,
"CommissionCNC": 0,
"CTNExchangeRate": null,
"MarkedForUpdate": false,
"ReceivedByGumar": false,
"ErrorSendToGumar": false,
"BankId": 2,
"ConsigneeId": 43214,
"ReExport": false,
"DCNumber": "7777777",
"CreatedOn": "2025-11-14T10:30:00Z",
"CreatedById": 1,
"CTN_Addresses": [
  {
    "Id": null,
    "QryAddressTypeId": 1,
    "Name": "EXPORTER COMPANY LTD",
    "Address": "123 Export Street",
    "City": "Export City",
    "CountryId": 10,
    "Email": "exporter@company.com",
    "Telephone": "244123456789",
    "NIFNumber": "1234567890"
  },
  {
    "Id": null,
    "QryAddressTypeId": 2,
```

```
"Name": "M.A.C.S AS.TEC.N. E IND",
"Address": "maculusso rua comandate",
"City": "kwenhan",
"CountryId": 4,
"Email": "contact@company.com",
"Telephone": "923798669",
"NIFNumber": "0000000458608"
}
],
"CTN_Goods": [
{
  "Id": null,
  "GoodsClassificationId": 16882,
  "IMOId": 2,
  "CargoTypeId": 1,
  "DescriptionGoods": "Electronic Components and Parts",
  "GrossWeightKG": 1000.0,
  "VolumeCBM": 11.0,
  "OceanFreight": 22.0,
  "ValueOfGoods": 33.0,
  "NumberOfPackages": 3
}
],
"CTN_Containers": [
{
  "Id": null,
  "ContainerTypeId": 31,
  "Number_of_Containers": 1,
  "ContainerNumbers": "MSCU1234567",
  "SealNumbers": "23234",
  "Groupage": false,
  "IsEmpty": false,
  "OwnedByShipper": false
}
],
"CTN_Tracking": [
{
  "Id": null,
  "SourceCountryId": 10,
  "SourcePortId": 80,
  "ETD": "2025-11-15T00:00:00.000Z",
  "TransportTypeId": 1,
  "VoyageNumber": "546894",
  "CarrierId": 789,
  "VesselId": 5163,
  "DestinationCountryId": 4,
  "DestinationPortId": 35,
  "ETA": "2025-12-09T00:00:00.000Z",
  "Sequence": 1
}
],
"CTN_RORO": [],
"CTN_Attachments": [
{
  "Id": null,
  "AttachmentGuid": "0da7461c-2155-4d20-a695-6b7463367327",
```

```

      "NameId": 4
    },
    {
      "Id": null,
      "AttachmentGuid": "1eb8572d-3266-5e31-b7a6-7c8574468438",
      "NameId": 1
    }
  ]
}

```

Sample JSON Response:

```

{
  "Id": 503808,
  "CTN_Reference_Number": "AO-CNT-503808-2025",
  "StatusId": 1,
  "BL_number": "vc568009iujh",
  "UniqueTradeNumber": "56789098765",
  "CreatedOn": "2025-11-14T10:30:00Z",
  "CreatedById": 1,
  "AddressesCreated": 2,
  "GoodsCreated": 1,
  "ContainersCreated": 1,
  "TrackingCreated": 1,
  "AttachmentsCreated": 2,
  "Message": "CTN certificate created successfully with all associated data"
}

```

5.2.5 Request Visa API (Certificate Issuance Submission)

Business Purpose: Submits completed CTN certificates for approval and visa issuance essential for finalizing the certificate approval process. Critical for triggering certificate review, validating completeness requirements, initiating approval workflow, and moving certificates from draft to approval status.

Endpoint: `POST /api/ctns/actions/requestvisa/{id}`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	Content-Length	Integer	M	Request body length (0 for this action)	0

Pre-submission Validation Requirements:

- **Shipper Address:** Complete shipper address information required
- **Forwarder Address:** Complete forwarder address information required
- **Attachments:** At least 1 attachment must be uploaded
- **CTN Data:** All mandatory CTN fields must be completed

UI Validation Rules:

- **REQVISA_UI_001:** CTN must be in "Edited" status to request visa
- **REQVISA_UI_002:** Display validation errors clearly to user
- **REQVISA_UI_003:** Confirm all requirements before submission

Business Validation Rules:

- **REQVISA_BV_001:** CTN must have complete address information
- **REQVISA_BV_002:** Required attachments must be uploaded
- **REQVISA_BV_003:** CTN must pass all business validations

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	rowsAffected	Integer	M	Number of affected records	-1
2	error	String - 1000	O	Validation error messages	""
3	newCtnId	Integer	M	CTN identifier (0 for existing)	0

Error Codes:

- **REQVISA_E001:** "Missing or incomplete Shipper Address" (HTTP 400)
- **REQVISA_E002:** "Missing or incomplete Forwarder Address" (HTTP 400)
- **REQVISA_E003:** "At least 1 attachment must be uploaded" (HTTP 400)
- **REQVISA_E004:** "CTN not found or not accessible" (HTTP 404)

Sample JSON Request:

```
POST /api/ctns/actions/requestvisa/503808
Content-Length: 0
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "rowsAffected": -1,
  "error": "",
  "newCtnId": 0
}
```

5.2.5.1 Document Upload API for Request Visa

Business Purpose: Enables SINTECE to upload documents specifically for the Request Visa process, providing document management capabilities separate from the general file upload API. Critical for visa application documentation, compliance verification, and regulatory submission requirements.

Endpoint: `POST /api/ctns/{ctnId}/documents/upload`

Request Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	ctnId	Integer	M	CTN identifier from URL path	503808
2	file	Binary	M	Document file (multipart/form-data)	[binary file data]
3	documentType	String - 50	M	Type of document being uploaded	BILL_OF_LADING
4	documentName	String - 256	M	Document name/title	Bill of Lading Original
5	description	String - 500	O	Document description	Original shipping document for visa application
6	isRequired	Boolean	M	Whether document is mandatory	true

UI Validation Rules:

- **DOCUP_UI_001:** File size must not exceed 25MB for visa documents
- **DOCUP_UI_002:** Document type must be from approved list
- **DOCUP_UI_003:** Document name required, max 256 characters
- **DOCUP_UI_004:** File types allowed: PDF, JPG, PNG, DOC, DOCX, XLS, XLSX

Business Validation Rules:

- **DOCUP_BV_001:** CTN must exist and be accessible
- **DOCUP_BV_002:** Document type must be valid for visa application
- **DOCUP_BV_003:** User must have permission to upload documents
- **DOCUP_BV_004:** Duplicate documents should be detected

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	documentId	String (UUID)	M	Unique document identifier	f47ac10b-58cc-4372-a567-0e02b2c3d479
2	documentName	String - 256	M	Uploaded document name	Bill of Lading Original
3	fileName	String - 256	M	Original file name	BOL_ABC123.pdf
4	fileSize	Integer	M	File size in bytes	1048576
5	mimeType	String - 100	M	File MIME type	application/pdf
6	downloadUrl	String - 500	M	URL to download	/api/ctns/503808/documents/f47ac10b-58cc-

				document	4372-a567-0e02b2c3d479/download
7	uploadDate	DateTime	M	Upload timestamp	2025-11-14T15:30:00Z
8	documentType	String - 50	M	Document type	BILL_OF_LADING
9	status	String - 20	M	Document status	UPLOADED
10	ctnId	Integer	M	Associated CTN ID	503808

Error Codes:

- **DOCUP_E001:** "CTN not found" (HTTP 404)
- **DOCUP_E002:** "Invalid document type" (HTTP 400)
- **DOCUP_E003:** "File size exceeds limit" (HTTP 413)
- **DOCUP_E004:** "Invalid file type" (HTTP 415)
- **DOCUP_E005:** "Document upload failed" (HTTP 500)

Sample Multipart Request:

```
POST /api/ctns/503808/documents/upload
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary7MA4YWxkTrZu0gW
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="file"; filename="BOL_ABC123.pdf"
Content-Type: application/pdf

[Binary PDF file content]
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="documentType"

BILL_OF_LADING
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="documentName"

Bill of Lading Original
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="description"

Original shipping document for visa application
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="isRequired"

true
-----WebKitFormBoundary7MA4YWxkTrZu0gW--
```

Sample JSON Response:

```
{
  "documentId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "documentName": "Bill of Lading Original",
```

```

"fileName": "BOL_ABC123.pdf",
"fileSize": 1048576,
"mimeType": "application/pdf",
"downloadUrl": "/api/ctns/503808/documents/f47ac10b-58cc-4372-a567-0e02b2c3d479/download",
"uploadDate": "2025-11-14T15:30:00Z",
"documentType": "BILL_OF_LADING",
"status": "UPLOADED",
"ctnId": 503808
}

```

5.2.5.2 Document Delete API for Request Visa

Business Purpose: Allows SINTECE to delete uploaded documents from CTN certificates before visa submission. Critical for document management, correcting upload mistakes, replacing outdated documents, and maintaining clean documentation sets.

Endpoint: `DELETE /api/ctns/{ctnId}/documents/{documentId}`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	ctnId	Integer	M	CTN identifier from URL path	503808
2	documentId	String (UUID)	M	Document identifier from URL path	f47ac10b-58cc-4372-a567-0e02b2c3d479
3	reason	String - 200	O	Deletion reason (in request body)	Incorrect document uploaded

UI Validation Rules:

- **DOCDEL_UI_001:** Confirm deletion with user before proceeding
- **DOCDEL_UI_002:** Display document details before deletion
- **DOCDEL_UI_003:** Reason for deletion optional but recommended

Business Validation Rules:

- **DOCDEL_BV_001:** CTN must exist and be accessible
- **DOCDEL_BV_002:** Document must exist and belong to CTN
- **DOCDEL_BV_003:** User must have permission to delete documents
- **DOCDEL_BV_004:** Cannot delete documents after visa submission
- **DOCDEL_BV_005:** Cannot delete required documents unless replacing

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	success	Boolean	M	Deletion success indicator	true
2	documentId	String (UUID)	M	Deleted document	f47ac10b-58cc-4372-a567-

				identifier	0e02b2c3d479
3	documentName	String - 256	M	Name of deleted document	Bill of Lading Original
4	deletedAt	DateTime	M	Deletion timestamp	2025-11-14T16:45:00Z
5	deletedBy	Integer	M	User who deleted document	13345
6	message	String - 200	M	Confirmation message	Document successfully deleted

Error Codes:

- **DOCDEL_E001:** "CTN not found" (HTTP 404)
- **DOCDEL_E002:** "Document not found" (HTTP 404)
- **DOCDEL_E003:** "Cannot delete after visa submission" (HTTP 409)
- **DOCDEL_E004:** "Cannot delete required document" (HTTP 409)
- **DOCDEL_E005:** "Insufficient permissions" (HTTP 403)

Sample JSON Request:

```
DELETE /api/ctns/503808/documents/f47ac10b-58cc-4372-a567-0e02b2c3d479
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

{
  "reason": "Incorrect document uploaded, need to replace with updated version"
}
```

Sample JSON Response:

```
{
  "success": true,
  "documentId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "documentName": "Bill of Lading Original",
  "deletedAt": "2025-11-14T16:45:00Z",
  "deletedBy": 13345,
  "message": "Document successfully deleted"
}
```

5.2.6 NIF Validation API (Angola Tax Registration Number)

Business Purpose: Validates NIF (Número de Identificação Fiscal) - Angola Tax Registration Number for importers, exporters, and consignees. Critical for ensuring tax compliance, preventing fraud, validating business entities, and meeting ARCLLA regulatory requirements for CNCA certificate processing.

Endpoint: `POST /api/validation/nif`

Request Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	nifNumber	String - 14	M	Angola Tax Registration Number (9-14 digits)	5000000000
2	entityName	String - 200	O	Name of entity for cross-validation	ABC Trading Company
3	entityType	String - 20	O	Type of entity (IMPORTER, EXPORTER, CONSIGNEE)	IMPORTER
4	validationType	String - 20	O	Validation type (STANDARD, ENHANCED)	STANDARD

UI Validation Rules:

- **NIF_UI_001:** NIF number mandatory, 9-14 alphanumeric characters
- **NIF_UI_002:** Entity name optional, max 200 characters
- **NIF_UI_003:** Entity type from predefined list only
- **NIF_UI_004:** Remove leading/trailing spaces from NIF

Business Validation Rules:

- **NIF_BV_001:** NIF must follow Angola tax authority format
- **NIF_BV_002:** NIF must be active and valid in ARCLA system
- **NIF_BV_003:** Entity name must match registered name if provided
- **NIF_BV_004:** Suspended or blacklisted NIFs cannot be used
- **NIF_BV_005:** NIF entity type must match certificate type (import/export)

Response Elements:

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	validationStatus	String - 20	M	Validation result (VALID, INVALID, SUSPENDED)	VALID
2	nifNumber	String - 14	M	Validated NIF number	5000000000
3	registeredName	String - 200	O	Official registered name from tax authority	ABC Trading Company Lda
4	entityStatus	String - 20	O	Entity status (ACTIVE, INACTIVE, SUSPENDED)	ACTIVE
5	registrationDate	Date	O	NIF registration date	2020-05-15
6	validationCode	String - 10	M	Validation result code	NIF_VALID
7	validationMessage	String - 500	M	Detailed validation message	NIF is valid and active

8	errorDetails	Array	O	Array of validation errors if any	[]
---	--------------	-------	---	-----------------------------------	----

Error Codes:

- **NIF_E001:** "Invalid NIF format" (HTTP 400)
- **NIF_E002:** "NIF not found in ARCCLA system" (HTTP 404)
- **NIF_E003:** "NIF suspended or blacklisted" (HTTP 403)
- **NIF_E004:** "Entity name mismatch" (HTTP 400)
- **NIF_E005:** "NIF validation service unavailable" (HTTP 503)
- **NIF_E006:** "Invalid entity type for certificate" (HTTP 400)

Sample JSON Request:

```
POST /api/validation/nif
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

{
  "nifNumber": "5000000000",
  "entityName": "ABC Trading Company",
  "entityType": "IMPORTER",
  "validationType": "STANDARD"
}
```

Sample JSON Response (Success):

```
{
  "validationStatus": "VALID",
  "nifNumber": "5000000000",
  "registeredName": "ABC Trading Company Lda",
  "entityStatus": "ACTIVE",
  "registrationDate": "2020-05-15",
  "validationCode": "NIF_VALID",
  "validationMessage": "NIF is valid and active in ARCCLA system",
  "errorDetails": []
}
```

Sample JSON Response (Error):

```
{
  "validationStatus": "INVALID",
  "nifNumber": "5000000000",
  "registeredName": null,
  "entityStatus": null,
  "registrationDate": null,
  "validationCode": "NIF_NOT_FOUND",
  "validationMessage": "NIF not found in ARCCLA tax registration system",
  "errorDetails": [
    {
      "field": "nifNumber",
      "code": "NIF_E002",
      "message": "The provided NIF does not exist in the Angola tax authority database"
    }
  ]
}
```

```
}  
]  
}
```

Integration Notes:

- NIF validation should be performed before CTN certificate creation
- Cached validation results valid for 24 hours
- Real-time validation with ARCCLA tax authority system
- Supports bulk NIF validation (future enhancement)

5.2.7 Invoice Download API**Important Note on Invoice Applicability:**

CRITICAL: Invoices are *ONLY* applicable for *IMPORT* profiles. Export profiles do *NOT* require invoice generation or download functionality. The invoice download API should only be called for import certificate types.

Business Purpose: Enables download of CNCA certificate invoices after certificate completion for import profiles only. Critical for financial record-keeping, customs documentation, and audit trail maintenance. **Not applicable for export certificates.**

Endpoint: `GET /api/invoices/{invoiceId}/download`

Request Elements:

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	invoiceId	String - 50	M	Unique invoice identifier from certificate	INV-2024-001234
2	format	String - 10	O	Invoice format (PDF, XML) - default PDF	PDF
3	certificateType	String - 20	O	Certificate type for validation (IMPORT/EXPORT)	IMPORT

Path Parameters:

- `invoiceId` : Unique invoice identifier (mandatory)

Query Parameters:

- `format` : Desired file format (optional, defaults to PDF)

UI Validation Rules:

- **INV_UI_001:** Invoice ID mandatory, alphanumeric with hyphens
- **INV_UI_002:** Format must be PDF or XML only
- **INV_UI_003:** Download only available for import profiles
- **INV_UI_004:** Display warning for export certificate requests

Business Validation Rules:

- **INV_BV_001:** Invoice must exist and be associated with authenticated user
- **INV_BV_002:** Certificate must be in "Approved" or "Issued" status
- **INV_BV_003:** User must have permission to access invoice

- **INV_BV_004:** Invoice download only allowed for IMPORT certificates
- **INV_BV_005:** Export certificates should not generate or download invoices

Response Elements:

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	fileName	String - 100	M	Generated file name	invoice_INV-2024-001234.pdf
2	fileSize	Number	M	File size in bytes	245678
3	contentType	String - 50	M	MIME type of file	application/pdf
4	downloadUrl	String - 500	O	Direct download URL (if applicable)	https://jul.adports.ae/downloads/...
5	invoiceData	Binary	M	Invoice file binary content (Base64 encoded)	[binary data]
6	generatedDate	Date	M	Invoice generation timestamp	2025-11-14T10:30:00Z

Error Codes:

- **INV_E001:** "Invoice not found" (HTTP 404)
- **INV_E002:** "Certificate not completed" (HTTP 400)
- **INV_E003:** "Access denied" (HTTP 403)
- **INV_E004:** "Invalid invoice format requested" (HTTP 400)
- **INV_E005:** "Export certificates do not support invoices" (HTTP 400)
- **INV_E006:** "Invoice generation in progress" (HTTP 202)

Sample API Request:

```
GET /api/invoices/INV-2024-001234/download?format=PDF
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "fileName": "invoice_INV-2024-001234.pdf",
  "fileSize": 245678,
  "contentType": "application/pdf",
  "downloadUrl": "https://jul.adports.ae/api/files/download/abc123xyz",
  "generatedDate": "2025-11-14T10:30:00Z",
  "invoiceData": "JVBERi0xLjQKJeLjz9MKMSAwIG9iago8PC9UeXB1L0NhdGFsb2cvUGFnZXMGMiAwIFI+..."
}
```

Invoice Applicability Rules:

Certificate Type	Invoice Required	Invoice Download Available	Notes
------------------	------------------	----------------------------	-------

IMPORT	✔ Yes	✔ Yes	Invoice generated after approval
EXPORT	✘ No	✘ No	Export certificates do not require invoices

Important Implementation Notes:

1. **Pre-Download Validation:** Always validate certificate type before allowing invoice download
2. **UI Display:** Do not show invoice download button/option for export certificates
3. **Error Handling:** Return specific error (INV_E005) when export certificate attempts invoice download
4. **Business Logic:** Invoice generation workflow should only trigger for import certificates
5. **Audit Trail:** Log all invoice download attempts with certificate type information

Integration Workflow:

1. Certificate approved by ARCCLA
2. For IMPORT certificates: Invoice generated automatically
3. Certificate becomes available for download
4. User downloads invoice via API or UI
5. For EXPORT certificates: Skip invoice generation entirely

5.3 CTN Related Entity APIs

CTN Related Entity APIs manage the specific entities and components that are directly associated with CTN certificate processing, including consignee information, attachment management, and tracking data.

5.3.1 Consignees API

Business Purpose: Manages consignee information essential for CTN certificate processing and customs compliance. Critical for identifying cargo recipients, validating company details, ensuring proper documentation, and enabling accurate certificate issuance with verified consignee data.

Endpoint: `GET /api/Consignees`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	NIFNumber
2	NIFNumber	String - 50	O	Filter by NIF number	123456789
3	CompanyName	String - 200	O	Filter by company name	ACME
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CONSIGNEE_UI_001:** NIF number format validation (9-14 digits)
- **CONSIGNEE_UI_002:** Company name minimum 3 characters
- **CONSIGNEE_UI_003:** Display with address for identification

Business Validation Rules:

- **CONSIGNEE_BV_001:** NIF number must be unique and valid
- **CONSIGNEE_BV_002:** Company must be registered and active
- **CONSIGNEE_BV_003:** Consignee must have valid import license

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique consignee identifier	1
2	NIFNumber	String - 50	M	Tax identification number	123456789
3	CompanyName	String - 200	M	Company/consignee name	ACME Import Company
4	ContactEmail	String - 100	O	Contact email address	contact@acme.ao
5	Address	String - 500	O	Company address	Rua da Independencia 123
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **CONSIGNEE_E001:** "Invalid NIF number format" (HTTP 400)
- **CONSIGNEE_E002:** "Consignee not found" (HTTP 404)
- **CONSIGNEE_E003:** "Consignee not authorized for import" (HTTP 422)

Sample JSON Request:

```
GET /api/Consignees?$sort=NIFNumber&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "NIFNumber": "123456789",
    "CompanyName": "ACME Import Company",
    "ContactEmail": "contact@acme.ao",
    "Address": "Rua da Independencia 123, Luanda",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.3.2 Attachment Names API

Business Purpose: Manages standardized attachment type classifications essential for CTN document management and compliance verification. Critical for organizing certificate documents, ensuring complete documentation, and maintaining audit trails for regulatory compliance.

Endpoint: `GET /api/AttachmentNames`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	AttachmentName_Desc
2	AttachmentName_Desc	String - 200	O	Filter by attachment name	Bill of Lading
3	DocumentType	String - 50	O	Filter by document type	Shipping
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **ATTACHMENT_UI_001:** Attachment name filter minimum 3 characters
- **ATTACHMENT_UI_002:** Group by document type for organization
- **ATTACHMENT_UI_003:** Display with required/optional indicator

Business Validation Rules:

- **ATTACHMENT_BV_001:** Some attachment types are mandatory
- **ATTACHMENT_BV_002:** File format restrictions apply by type
- **ATTACHMENT_BV_003:** Attachment size limits vary by type

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique attachment type identifier	1
2	AttachmentName_Desc	String - 200	M	Attachment type description	Bill of Lading
3	DocumentType	String - 50	O	Document category	Shipping
4	IsRequired	Boolean	O	Mandatory attachment indicator	true
5	MaxFileSizeMB	Integer	O	Maximum file size in MB	10
6	Active	Boolean	M	Active status indicator	true

Error Codes:

- **ATTACHMENT_E001:** "Invalid attachment parameters" (HTTP 400)
- **ATTACHMENT_E002:** "Attachment type not found" (HTTP 404)
- **ATTACHMENT_E003:** "File exceeds size limit" (HTTP 413)

Sample JSON Request:

```
GET /api/AttachmentNames?$sort=AttachmentName_Desc&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "AttachmentName_Desc": "Bill of Lading",
    "DocumentType": "Shipping",
    "IsRequired": true,
    "MaxFileSizeMB": 10,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "AttachmentName_Desc": "Commercial Invoice",
    "DocumentType": "Financial",
    "IsRequired": true,
    "MaxFileSizeMB": 5,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.3.3 CTN Tracking API

Business Purpose: Manages CTN shipment tracking information essential for logistics monitoring and cargo visibility. Critical for providing real-time shipment status, managing transport schedules, and enabling proactive logistics coordination throughout the shipping lifecycle.

Endpoint: `GET /api/ctnTracking`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand parameter	CTN, DestinationPort, SourcePort, TransportType
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier	503808

(alternative)

UI Validation Rules:

- **TRACKING_UI_001:** CTN ID required for tracking lookup
- **TRACKING_UI_002:** Display tracking timeline visualization
- **TRACKING_UI_003:** Show estimated vs actual times with status

Business Validation Rules:

- **TRACKING_BV_001:** CTN must exist and be active
- **TRACKING_BV_002:** Tracking dates must be logical sequence
- **TRACKING_BV_003:** Vessel and voyage information must be consistent

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Tracking record ID	71898
2	CTNId	Integer	M	Related CTN ID	503808
3	SourcePortId	Integer	O	Source port identifier	150
4	DestinationPortId	Integer	O	Destination port identifier	275
5	TransportTypeId	Integer	M	Transport type identifier	1
6	ETD	DateTime	O	Estimated time of departure	2023-12-15T10:00:00Z
7	ETA	DateTime	O	Estimated time of arrival	2023-12-20T15:30:00Z
8	ATD	DateTime	O	Actual time of departure	2023-12-15T11:15:00Z
9	ATA	DateTime	O	Actual time of arrival	2023-12-20T14:45:00Z
10	VesselId	Integer	O	Vessel identifier	5163
11	VoyageNumber	String - 50	O	Voyage number	546894

Error Codes:

- **TRACKING_E001:** "Invalid CTN ID" (HTTP 400)
- **TRACKING_E002:** "Tracking information not found" (HTTP 404)
- **TRACKING_E003:** "Tracking data inconsistent" (HTTP 422)

Sample JSON Request:

```
GET /api/ctnTracking?$expand=CTN, DestinationPort, SourcePort, TransportType&ctnid=503808
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 71898,
    "CTNId": 503808,
    "SourcePortId": 80,
    "DestinationPortId": 35,
    "TransportTypeId": 1,
    "ETD": "2025-11-12T00:00:00.000Z",
    "ETA": "2025-12-09T00:00:00.000Z",
    "ATD": null,
    "ATA": null,
    "VesselId": 5163,
    "VoyageNumber": "546894",
    "CreatedOn": "2025-11-12T18:49:00Z",
    "CreatedById": 13345,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.3.4 CTN Containers API**Endpoint:** GET /api/ctnContainers**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	ContainerType
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Container record ID	510046
2	CTNId	Integer	M	Related CTN ID	503808
3	ContainerNumber	String - 20	M	Container number	MSCU1234567
4	ContainerTypeId	Integer	M	Container type ID	1
5	SealNumbers	String - 200	O	Container seal numbers	SEAL123,SEAL456
6	TareWeightKG	Decimal	M	Tare weight in kg	2300.00

7	GrossWeightKG	Decimal	M	Gross weight in kg	18500.00
8	NetWeightKG	Decimal	M	Net weight in kg	16200.00

POST /api/ctnContainers - Create Container Record**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	ContainerNumber	String - 20	M	Container number	MSCU1234567
3	ContainerTypeId	Integer	M	Container type identifier	1
4	SealNumbers	String - 200	O	Container seal numbers	SEAL123,SEAL456
5	TareWeightKG	Decimal	M	Tare weight in kg	2300.00
6	GrossWeightKG	Decimal	M	Gross weight in kg	18500.00

GET /api/ctnContainers/new - Container Template**Response Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Template ID (null for new)	null
2	CTNId	Integer	O	CTN ID (to be set)	null
3	ContainerNumber	String - 20	O	Empty container number	""
4	ContainerTypeId	Integer	O	Default container type	1
5	TareWeightKG	Decimal	O	Default tare weight	2300.00

Business Validation Rules:

- **CONTAINER_BV_001:** Container number must follow ISO 6346 standard
- **CONTAINER_BV_002:** Container type must be valid and active
- **CONTAINER_BV_003:** Seal numbers must be unique per container

Error Codes:

- **CONTAINER_E001:** "Invalid container number pattern" (HTTP 400)
- **CONTAINER_E002:** "Container already exists" (HTTP 409)
- **CONTAINER_E003:** "Invalid container type" (HTTP 422)

Sample JSON Request (GET):

```
GET /api/ctnContainers?$expand=ContainerType&ctn=503808&ctnid=503808
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response (GET):

```
[]
```

Sample JSON Request (POST):

```
POST /api/ctnContainers
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

{
  "Id": null,
  "CTNId": 503808,
  "ContainerTypeId": 31,
  "Number_of_Containers": 1,
  "ContainerNumbers": "MSCU1234567",
  "SealNumbers": "23234",
  "Groupage": false,
  "ContainerType": null,
  "CTN": null,
  "IsEmpty": false,
  "OwnedByShipper": false
}
```

Sample JSON Response (POST Success):

```
{
  "Id": 510046,
  "CTNId": 503808,
  "ContainerTypeId": 31,
  "Number_of_Containers": 1,
  "ContainerNumbers": "MSCU1234567",
  "SealNumbers": "23234",
  "Groupage": false,
  "ContainerType": null,
  "CTN": null,
  "IsEmpty": false,
  "OwnedByShipper": false
}
```

Sample JSON Response (GET /new Template):

```
{
  "Id": null,
  "CTNId": null,

```

```

    "ContainerTypeId": null,
    "Number_of_Containers": 1,
    "ContainerNumbers": null,
    "SealNumbers": null,
    "Groupage": false,
    "ContainerType": null,
    "CTN": null,
    "IsEmpty": false,
    "OwnedByShipper": false
  }

```

5.3.5 CTN Addresses API

Endpoint: GET /api/ctnAddresses

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	QryAddressType,Country1
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Address record ID	426334
2	CTNId	Integer	M	Related CTN ID	503808
3	AddressTypeId	Integer	M	Address type ID	1
4	CompanyName	String - 200	M	Company name	ABC Trading LLC
5	ContactPerson	String - 100	O	Contact person name	John Smith
6	Address1	String - 200	M	Address line 1	123 Business Street
7	Address2	String - 200	O	Address line 2	Suite 456
8	City	String - 100	M	City	Dubai
9	CountryId	Integer	M	Country identifier	4
10	PostalCode	String - 20	O	Postal/ZIP code	12345
11	Phone	String - 50	O	Phone number	+971-4-1234567

12	Email	String - 200	O	Email address	contact@abc-trading.com
----	-------	--------------	---	---------------	--

5.3.6 CTN Charges API

Endpoint: GET /api/ctnCharges

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	Charge,ctn.view_currency
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Charge record ID	12345
2	CTNId	Integer	M	Related CTN ID	503808
3	ChargeId	Integer	M	Charge type ID	1
4	Amount	Decimal	M	Charge amount	150.00
5	CurrencyId	Integer	M	Currency identifier	2
6	Description	String - 200	O	Charge description	Certificate Processing Fee
7	IsSystem	Boolean	M	System-generated charge flag	true

5.3.7 User Communication API

Endpoint: GET /api/userCommunication

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 100	O	OData expand parameter	modifiedby
2	\$sort	String - 100	O	OData sort parameter	-modifiedon
3	ctn	Integer	M	CTN identifier	503808
4	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Communication record ID	98765
2	CTNId	Integer	M	Related CTN ID	503808
3	Subject	String - 200	M	Communication subject	Document Review Required
4	Message	String - 2000	M	Communication message	Please review attached documents
5	CreatedOn	DateTime	M	Creation timestamp	2023-11-12T16:45:00Z
6	CreatedById	Integer	M	User who created the message	13345
7	IsInternal	Boolean	M	Internal communication flag	false

POST /api/userCommunication - Create Communication**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	Subject	String - 200	M	Communication subject	Document Review Required
3	Message	String - 2000	M	Communication message	Please review attached documents
4	IsInternal	Boolean	M	Internal communication flag	false
5	RecipientIds	Array	O	Array of recipient user IDs	[13345, 13346]

5.4 File Management APIs**5.4.1 File Upload API****Endpoint:** `POST /api/fileupload`**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	file	Binary	M	File upload (multipart/form-data)	[binary file data]
2	fileName	String - 256	M	Original file name	Bill_of_Lading.pdf
3	description	String - 500	O	File description	Original shipping document

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fileId	String (UUID)	M	Unique file identifier	0da7461c-2155-4d20-a695-6b7463367327
2	fileName	String - 256	M	Uploaded file name	Bill_of_Lading.pdf
3	fileSize	Integer	M	File size in bytes	245760
4	contentType	String - 100	M	File MIME type	application/pdf
5	uploadDate	DateTime	M	Upload timestamp	2023-11-12T14:30:00Z

UI Validation Rules:

- **FILEUP_UI_001:** File size must not exceed 10MB
- **FILEUP_UI_002:** Only approved file types allowed (PDF, JPG, PNG, DOC, DOCX)
- **FILEUP_UI_003:** File name must not contain special characters

Business Validation Rules:

- **FILEUP_BV_001:** File must not be corrupted or malicious
- **FILEUP_BV_002:** Duplicate files should be detected and warned
- **FILEUP_BV_003:** User must have permission to upload files

Error Codes:

- **FILEUP_E001:** "File size exceeds maximum limit" (HTTP 413)
- **FILEUP_E002:** "File type not supported" (HTTP 415)
- **FILEUP_E003:** "File upload failed" (HTTP 500)

Sample Multipart Request:

```
POST /api/fileupload
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary7MA4YWxkTrZu0gW
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="file"; filename="Bill_of_Lading.pdf"
Content-Type: application/pdf

[Binary PDF file content]
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="fileName"

Bill_of_Lading.pdf
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="description"

Original shipping document for CTN certificate
-----WebKitFormBoundary7MA4YWxkTrZu0gW--
```

Sample JSON Response:

```
{
  "fileId": "0da7461c-2155-4d20-a695-6b7463367327",
  "fileName": "Bill_of_Lading.pdf",
  "fileSize": 245760,
  "mimeType": "application/pdf",
  "uploadDate": "2023-11-12T14:30:00Z"
}
```

GET /api/fileupload/{fileId} - Download File**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fileId	String (UUID)	M	File identifier from URL path	0da7461c-2155-4d20-a695-6b7463367327

Sample Download Request:

```
GET /api/fileupload/0da7461c-2155-4d20-a695-6b7463367327
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	contentStream	Base64 Binary	M	Base64 encoded file content	JVBERi0xLjQKJeLjz9MK...
2	fileName	String - 256	M	Original file name	Bill_of_Lading.pdf
3	mimeType	String - 100	M	File MIME type	application/pdf
4	fileSize	Integer	M	File size in bytes	245760

Sample Download Response:

```
{
  "contentStream":
    "JVBERi0xLjQKJeLjz9MKNSAwIG9iago8PC9MZW5ndGggNzggL0ZpbHRlciAvRm9udGVzZWNvZGU+PnN0cmVhbQ==",
  "fileName": "Bill_of_Lading.pdf",
  "mimeType": "application/pdf",
  "fileSize": 245760
}
```

5.6 Specialized CTN APIs

5.6.1 Parent CTN Relationships API

Endpoint: GET /api/Ctns/GetAllowedParentCtns

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	ctnId	Integer	M	Current CTN identifier	503808
2	\$sort	String - 100	O	OData sort parameter	BL_number
3	BL_number	String - 50	O	Filter by Bill of Lading number	MSC123456

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Parent CTN ID	503800
2	CTN_Reference_Number	String - 50	M	Parent CTN reference	AO-CNT-503800-2023
3	BL_number	String - 50	M	Parent Bill of Lading	MSC123456
4	StatusId	Integer	M	Parent CTN status	2
5	CanBeParent	Boolean	M	Eligibility as parent	true

5.6.2 CTN Export API

Endpoint: GET /api/ctns/export

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand parameter	CargoType,Status,Visum_Agent
2	\$export_format	String - 10	M	Export format	csv
3	\$sort	String - 100	O	OData sort parameter	-ModifiedOn
4	\$top	Integer	O	Number of records	10
5	\$filter	String - 1000	O	OData filter expression	StatusId eq 2

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
-------	------------	------------------	-----------	-------------------------	--------------

1	Content	Text/CSV	M	CSV formatted CTN data	"Id","CTN_Reference_Number"...
2	ContentType	String - 50	M	Response content type	text/csv
3	FileName	String - 100	M	Suggested file name	CTN_Export_20231112.csv

6. Enhanced API Capabilities ⚡

6.1 Amendment APIs

Amendment APIs provide controlled modification capabilities for existing certificates with ARCCLA approval workflow.

6.1.1 Amendment Request Creation API

Endpoint: `POST /api/CTNs/{id}/amendments`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	requestReason	String - 500	M	Reason for amendment request	Incorrect vessel name provided
3	businessJustification	String - 1000	O	Detailed business justification	Customer informed of vessel change after booking
4	requestedChanges	Array	M	Array of field changes requested	[field change objects]
5	urgencyLevel	String - 20	O	Urgency level (Normal, High, Critical)	Normal

Requested Changes Array Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fieldPath	String - 100	M	JSON path to field being changed	VesselName
2	fieldName	String - 100	M	Human-readable field name	Vessel Name
3	oldValue	String - 500	M	Current field value	MSC MEDITERRANEAN
4	newValue	String - 500	M	Proposed new field value	MAERSK ESSEX
5	changeReason	String - 200	M	Specific reason for this field change	Vessel change notification from carrier

Sample JSON Request:

```
{
  "requestReason": "Incorrect vessel name provided",
```

```
"businessJustification": "Customer informed of vessel change after booking confirmation",
"urgencyLevel": "Normal",
"requestedChanges": [
  {
    "fieldPath": "VesselName",
    "fieldName": "Vessel Name",
    "oldValue": "MSC MEDITERRANEAN",
    "newValue": "MAERSK ESSEX",
    "changeReason": "Vessel change notification from carrier"
  },
  {
    "fieldPath": "ETA",
    "fieldName": "Estimated Time of Arrival",
    "oldValue": "2023-12-20T15:30:00Z",
    "newValue": "2023-12-22T10:00:00Z",
    "changeReason": "Updated ETA due to vessel change"
  }
]
}
```

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Unique amendment identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String - 50	M	Sequential amendment number	AMD-503808-001
3	status	String - 20	M	Amendment status	Pending
4	requestDate	DateTime	M	Amendment request timestamp	2025-11-13T10:45:00Z
5	estimatedFee	Decimal	O	Estimated amendment fee	25.00

Sample JSON Response:

```
{
  "amendmentId": "a1b2c3d4-e5f6-7890-abcd-1234567890ef",
  "amendmentNumber": "AMD-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T10:45:00Z",
  "estimatedFee": 25.00,
  "currency": "USD",
  "submittedBy": 13345,
  "submittedByRole": "CustomsBroker"
}
```

6.1.2 Amendment Status API

Endpoint: `GET /api/amendments/{amendmentId}/status`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier from URL path	a1b2c3d4-e5f6-7890-abcd-1234567890ef

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String - 50	M	Amendment number	AMD-503808-001
3	status	String - 20	M	Current amendment status	UnderReview
4	assignedTo	String - 100	O	ARCCLA broker assigned	John.Smith@arccla.ae
5	lastUpdated	DateTime	M	Last status update timestamp	2025-11-13T14:30:00Z
6	statusHistory	Array	O	Amendment status history	[status history objects]

6.1.3 Amendment Approval API

Endpoint: `PUT /api/amendments/{amendmentId}/approve`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier from URL path	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	decision	String - 20	M	Approval decision (Approved/Rejected)	Approved
3	decisionReason	String - 1000	O	Reason for approval decision	Valid vessel change confirmed

4	reviewComments	String - 2000	O	Additional reviewer comments	Amendment approved without issues
5	amendmentFee	Decimal	O	Final amendment fee	25.00

6.2 Cancellation APIs

Cancellation APIs provide controlled termination of certificate requests with financial impact handling.

6.2.1 Cancellation Request Creation API

Endpoint: `POST /api/CTNs/{id}/cancellations`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	reasonCode	String - 20	M	Standardized cancellation reason code	SHIPMENT_CANCELLED
3	reasonDescription	String - 500	M	Detailed cancellation reason	Shipment cancelled by customer
4	additionalDetails	String - 1000	O	Additional cancellation details	Customer business closure
5	urgencyLevel	String - 20	O	Cancellation urgency (Normal, High, Critical)	High

Sample JSON Request:

```
{
  "reasonCode": "SHIPMENT_CANCELLED",
  "reasonDescription": "Shipment cancelled by customer due to business closure",
  "additionalDetails": "Customer has permanently ceased operations and requires immediate cancellation",
  "urgencyLevel": "High"
}
```

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Unique cancellation identifier	x1y2z3a4-b5c6-7890-defg-9876543210ab
2	cancellationNumber	String - 50	M	Sequential cancellation number	CAN-503808-001

3	status	String - 20	M	Cancellation status	Pending
4	requestDate	DateTime	M	Cancellation request timestamp	2025-11-13T11:30:00Z
5	estimatedRefund	Decimal	O	Estimated refund amount	125.00

Sample JSON Response:

```
{
  "cancellationId": "x1y2z3a4-b5c6-7890-defg-9876543210ab",
  "cancellationNumber": "CAN-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T11:30:00Z",
  "estimatedRefund": 125.00,
  "refundCurrency": "USD",
  "submittedBy": 13345,
  "submittedByRole": "CustomsBroker"
}
```

6.2.2 Cancellation Status API**Endpoint:** GET /api/cancellations/{cancellationId}/status**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Cancellation identifier from URL path	x1y2z3a4-b5c6-7890-defg-9876543210ab

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Cancellation identifier	x1y2z3a4-b5c6-7890-defg-9876543210ab
2	cancellationNumber	String - 50	M	Cancellation number	CAN-503808-001
3	status	String - 20	M	Current cancellation status	UnderReview
4	assignedTo	String - 100	O	ARCCLA broker assigned	Maria.Santos@arccla.ao
5	refundStatus	String - 20	O	Refund processing	Pending

				status	
6	lastUpdated	DateTime	M	Last status update timestamp	2025-11-13T15:45:00Z

6.3 Approval Workflow APIs

Approval workflow APIs enable ARCCLA brokers to review and process amendment/cancellation requests.

6.3.1 Pending Requests API

Endpoint: GET /api/approvals/pending

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	requestType	String - 20	O	Filter by request type (amendments/cancellations)	amendments
2	priority	String - 20	O	Filter by priority level	High
3	assignedTo	String - 100	O	Filter by assigned broker	current_user
4	\$top	Integer	O	Number of records to return	20
5	\$skip	Integer	O	Number of records to skip	0

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	requestId	String (UUID)	M	Amendment/Cancellation identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	requestType	String - 20	M	Type of request (Amendment/Cancellation)	Amendment
3	requestNumber	String - 50	M	Request number	AMD-503808-001
4	certificateId	String - 50	M	Related certificate ID	503808
5	submittedDate	DateTime	M	Request submission date	2025-11-13T10:45:00Z
6	priority	String - 20	M	Request priority	Normal
7	submittedBy	String - 100	M	User who submitted request	CustomsBroker123

6.4 Status Polling APIs

Status polling APIs provide real-time status updates with eligibility flags for allowed operations.

6.4.1 Enhanced Certificate Status API

Endpoint: `GET /api/CTNs/{id}/enhanced-status`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	includeHistory	Boolean	O	Include status change history	true
3	includeFlags	Boolean	O	Include eligibility flags	true

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	certificateId	Integer	M	Certificate identifier	503808
2	currentStatus	String - 20	M	Current certificate status	Approved
3	statusId	Integer	M	Status identifier	3
4	canAmend	Boolean	M	Amendment eligibility flag	true
5	canCancel	Boolean	M	Cancellation eligibility flag	true
6	lastAmendmentDate	DateTime	O	Date of last amendment	2025-11-10T14:20:00Z
7	pendingAmendments	Integer	M	Number of pending amendments	0
8	pendingCancellations	Integer	M	Number of pending cancellations	0
9	eligibilityReasons	Object	O	Detailed eligibility explanations	{...}

Sample JSON Response:

```
{
  "certificateId": 503808,
  "currentStatus": "Approved",
  "statusId": 3,
  "canAmend": true,
  "canCancel": true,
  "lastStatusUpdate": "2025-11-13T09:30:00Z",
}
```

```

    "lastAmendmentDate": null,
    "pendingAmendments": 0,
    "pendingCancellations": 0,
    "eligibilityReasons": {
      "amendmentEligibility": "Certificate is in approved status and within amendment timeframe",
      "cancellationEligibility": "Certificate not yet issued, cancellation allowed with full refund"
    },
    "businessRules": {
      "amendmentTimeLimit": "24 hours after approval",
      "cancellationTimeLimit": "48 hours after approval",
      "maxAmendmentsAllowed": 3
    }
  }
}

```

6.4.2 Status Retrieval API (Polling-Based Status Updates)

Business Purpose: Enables JUL system to retrieve current status of CNCA certificates by polling SINTECE system. Replaces webhook-based notifications with pull-based status updates, providing reliable status synchronization, eliminating callback URL management, and supporting firewall-friendly integration patterns.

Endpoint: `GET /api/ctns/{ctnId}/status`

Request Elements:

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	ctnId	Integer	M	CTN certificate identifier (path parameter)	503808
2	includeHistory	Boolean	O	Include status change history (query param)	false
3	includeDetails	Boolean	O	Include detailed status information (query param)	true

Path Parameters:

- `ctnId` : CTN certificate identifier (mandatory)

Query Parameters:

- `includeHistory` : Set to true to get full status change history
- `includeDetails` : Set to true to get detailed status information

UI Validation Rules:

- STATUS_UI_001:** CTN ID mandatory, positive integer
- STATUS_UI_002:** Boolean parameters only accept true/false
- STATUS_UI_003:** Display loading state during status retrieval

Business Validation Rules:

- STATUS_BV_001:** CTN must exist in system
- STATUS_BV_002:** User must have permission to view CTN status
- STATUS_BV_003:** Status must be current (not stale data)

Response Elements:

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	ctnId	Integer	M	CTN certificate identifier	503808
2	ctnReferenceNumber	String - 50	M	System generated CTN reference	170543
3	currentStatus	String - 50	M	Current certificate status	Approved
4	statusId	Integer	M	Status identifier	3
5	statusUpdatedAt	DateTime	M	Last status update timestamp	2025-11-14T10:30:00Z
6	statusUpdatedBy	String - 100	O	User who updated status	ARCCLA_Broker_123
7	canAmend	Boolean	M	Amendment eligibility flag	true
8	canCancel	Boolean	M	Cancellation eligibility flag	false
9	statusMessage	String - 500	O	Status description or notes	Certificate approved by ARCCLA
10	nextAction	String - 200	O	Recommended next action	Process payment and download certificate
11	statusHistory	Array	O	Status change history (if requested)	[...]

Status History Element (when includeHistory=true):

S. No	Attributes	Data Type - Length	Condition (M/O)	Format/Derivation logic for fields	Data Example
1	statusId	Integer	M	Status identifier	2
2	statusName	String - 50	M	Status name	Submitted
3	changedAt	DateTime	M	Status change timestamp	2025-11-13T14:20:00Z
4	changedBy	String - 100	O	User who changed status	Customs_Broker_456
5	remarks	String - 500	O	Status change remarks	Submitted for ARCCLA review

Error Codes:

- **STATUS_E001:** "CTN not found" (HTTP 404)

- **STATUS_E002:** "Access denied" (HTTP 403)
- **STATUS_E003:** "Invalid CTN ID format" (HTTP 400)
- **STATUS_E004:** "Status service temporarily unavailable" (HTTP 503)

Sample API Request:

```
GET /api/ctns/503808/status?includeHistory=true&includeDetails=true
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "ctnId": 503808,
  "ctnReferenceNumber": "170543",
  "currentStatus": "Approved",
  "statusId": 3,
  "statusUpdatedAt": "2025-11-14T10:30:00Z",
  "statusUpdatedBy": "ARCCLA_Broker_123",
  "canAmend": true,
  "canCancel": false,
  "statusMessage": "Certificate approved by ARCCLA. Ready for payment processing.",
  "nextAction": "Process payment and download certificate",
  "statusHistory": [
    {
      "statusId": 1,
      "statusName": "Created",
      "changedAt": "2025-11-12T18:49:30Z",
      "changedBy": "Customs_Broker_456",
      "remarks": "Initial CTN creation"
    },
    {
      "statusId": 2,
      "statusName": "Submitted",
      "changedAt": "2025-11-13T14:20:00Z",
      "changedBy": "Customs_Broker_456",
      "remarks": "Submitted for ARCCLA review"
    },
    {
      "statusId": 3,
      "statusName": "Approved",
      "changedAt": "2025-11-14T10:30:00Z",
      "changedBy": "ARCCLA_Broker_123",
      "remarks": "Certificate approved by ARCCLA"
    }
  ]
}
```

Polling Strategy Recommendations:

Certificate Status	Recommended Polling Interval	Notes

Created	No polling needed	Local status
Submitted	Every 5 minutes	Active review period
UnderReview	Every 10 minutes	ARCCLA processing
PendingInfo	Every 30 minutes	Waiting for trader response
Approved	Every 2 minutes	Payment processing
Issued	No polling needed	Final status
Cancelled	No polling needed	Final status

Implementation Guidelines:

1. **Efficient Polling:** Use appropriate intervals based on certificate status
2. **Rate Limiting:** Respect API rate limits (max 120 requests/minute per user)
3. **Caching:** Cache status responses for 30 seconds to reduce API calls
4. **Error Handling:** Implement exponential backoff for failed polling attempts
5. **User Experience:** Show real-time updates in UI based on polling results
6. **Background Jobs:** Use scheduled jobs for polling instead of continuous loops
7. **Notification:** Trigger user notifications when status changes detected

Advantages Over Webhooks:

-  No callback URL configuration required
 -  Firewall-friendly (outbound requests only)
 -  Better control over update frequency
 -  Simpler security model (no webhook signature validation)
 -  Easier error handling and retry logic
 -  No missed notifications due to network issues
-

7. API Implementation Summary

6.1 Complete CNCA Certificate API Coverage

The JUL-SINTECE Integration Control Document provides comprehensive API documentation covering the complete CNCA certificate lifecycle with section-by-section submission capabilities.

API Coverage Statistics:

- **Total APIs Documented:** 31 comprehensive APIs
- **Master Data APIs:** 15 essential reference data endpoints
- **Certificate Management APIs:** 11 core certificate workflow endpoints
- **CTN Related Entity APIs:** 5 supporting data endpoints
- **JSON Coverage:** 100% of APIs include production-ready request/response samples

Section-by-Section Certificate Submission APIs:

API Category	Endpoint	Purpose	Production Data
Core Creation	POST /api/ctns	Create initial CTN record	✓ CTN 503808
Address Management	POST /api/ctnAddresses	Add shipper/consignee/forwarder details	✓ Live addresses
Cargo Details	POST /api/ctnGoods	Add goods classifications and values	✓ Cargo data
Container Info	POST /api/ctnContainers	Add container types and numbers	✓ Container specs
Transport Routes	POST /api/ctnTracking	Add vessel and port information	✓ Route data
Document Attachment	POST /api/ctnAttachments	Link supporting documents	✓ File links
Final Submission	POST /api/ctns/actions/requestvisa/{id}	Submit for approval	✓ Visa requests

Key Integration Benefits:

1. **Progressive Data Entry:** Section-by-section approach improves user experience and data quality
2. **Comprehensive Validation:** Field-level and business rule validation at each step
3. **Production Ready:** All APIs tested with real SINTECE system data
4. **Complete Workflow:** From initial creation to final certificate approval
5. **Error Handling:** Detailed error codes and validation messages
6. **Audit Trail:** Complete tracking of certificate submission process

Workflow Completeness:

-  **Initial Creation:** Basic CTN record with core information
-  **Data Enhancement:** Section-wise addition of detailed information
-  **Document Management:** File upload and attachment linking
-  **Validation & Submission:** Complete data validation before approval request
-  **Status Tracking:** Real-time status monitoring throughout process

-  **Amendment Support:** Post-issuance certificate modification capabilities
-  **Cancellation Support:** Certificate cancellation workflow

6.2 Implementation Readiness

Development Team Benefits:

- Complete API specifications with production JSON samples
- Comprehensive error code documentation
- Business rule validation requirements
- Real system response examples for testing
- Step-by-step integration guidance

Business Process Coverage:

- End-to-end CNCA certificate lifecycle
 - Section-by-section submission for improved UX
 - Complete supporting document management
 - Automated validation and approval workflows
 - Real-time status tracking and notifications
-

8. Validation Framework

7.1 Field-Level Validation Rules

Text Field Validations:

- **TXT_001:** Required fields cannot be null or empty
- **TXT_002:** Text fields must not exceed maximum length
- **TXT_003:** Special characters properly escaped for security

Numeric Field Validations:

- **NUM_001:** Numeric fields must contain valid numbers
- **NUM_002:** Currency amounts non-negative with max 2 decimal places
- **NUM_003:** Integer IDs must be positive

Date/Time Validations:

- **DT_001:** All dates in ISO 8601 format (YYYY-MM-DDTHH:mm:ssZ)
- **DT_002:** ETD must be before or equal to ETA
- **DT_003:** Creation dates cannot be in future

7.2 Business Logic Validation

Trade Compliance:

- **TC_001:** Angola must be origin or destination for CNCA certificates
- **TC_002:** Sanctioned countries require special compliance screening
- **TC_003:** High-value shipments (>\$50,000) require enhanced documentation

Financial Validations:

- **FIN_001:** Invoice value must match bill of lading value ($\pm 5\%$ tolerance)
- **FIN_002:** Currency conversion rates current (within 24 hours)
- **FIN_003:** Payment terms consistent with Incoterms selected

Workflow State Validations:

- **ST_001:** Only valid status transitions allowed per business rules
- **ST_002:** Required actions completed before status advancement
- **ST_003:** Role-based permissions validated for status changes

7.3 Error Handling Standards

Error Response Structure:

```
{
  "error": {
    "code": "CTN_V001",
    "message": "Bill of Lading number must be unique",
    "severity": "ERROR",
    "field": "BL_number",
    "timestamp": "2025-11-13T16:45:00Z"
  }
}
```

Performance Standards:

- **PERF_001:** Field validations complete within 100ms
- **PERF_002:** Business validations complete within 500ms
- **PERF_003:** External validations complete within 2 seconds

8.3 Authentication and Authorization

The JUL-SINTECE integration uses Keycloak as the primary Single Sign-On (SSO) solution for centralized authentication and authorization management. Keycloak provides enterprise-grade identity and access management capabilities, ensuring secure and seamless authentication across both JUL and SINTECE systems.

8.3.1 SSO Solution - Keycloak

Overview: The JUL-SINTECE integration uses Keycloak as the primary Single Sign-On (SSO) solution for centralized authentication and authorization management. Keycloak provides enterprise-grade identity and access management capabilities, ensuring secure and seamless authentication across both JUL and SINTECE systems.

Key Features:

- Centralized user authentication and session management
- Single Sign-On (SSO) across multiple applications
- Token-based authentication with JWT
- Role-based access control (RBAC)
- Multi-factor authentication (MFA) support
- User federation and identity brokering

8.3.2 Supported Authentication Protocols

The client authentication system has the capability to integrate with Keycloak using the following protocols:

Protocol	Description	Use Case
OpenID Connect	Authentication layer built on OAuth 2.0	Recommended for user authentication and SSO
OAuth 2.0	Authorization framework for delegated access	API authorization and service-to-service communication
SAML 2.0	XML-based authentication protocol	Enterprise SSO integration (if required)

Primary Protocol: OpenID Connect (OIDC) is the recommended protocol for JUL-SINTECE integration.

8.3.3 Client Integration Capabilities

Current System Capabilities:

- JWT token support and validation
- Token-based authentication flow
- Support for Bearer token authentication
- Client credentials flow for service-to-service communication
- Authorization code flow for user authentication

Integration Approach:

1. **Client Registration:** Register JUL application in Keycloak realm
2. **Client Credentials:** Obtain client ID and client secret
3. **Token Endpoint:** Configure token endpoint for authentication

4. **User Info Endpoint:** Configure user information retrieval
5. **Logout Endpoint:** Implement proper logout and session management

8.3.4 JWT Token Management

Token Format and Structure: The system uses JSON Web Tokens (JWT) for secure authentication. All API requests must include a valid JWT token in the Authorization header.

Token Request:

```
POST /auth/realms/{realm}/protocol/openid-connect/token
Content-Type: application/x-www-form-urlencoded

grant_type=password&
client_id=jul-client&
client_secret={client_secret}&
username={username}&
password={password}
```

Token Response:

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expires_in": 3600,
  "refresh_expires_in": 1800,
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "not-before-policy": 0,
  "session_state": "d2d2b3b4-4f5e-4f3e-9c1a-5f6e7d8e9f0a",
  "scope": "profile email"
}
```

Token Usage in API Requests: All API requests must include the access token in the Authorization header:

```
GET /api/ctns
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Token Validation:

- Tokens are validated on every API request
- Signature verification using public key from Keycloak
- Expiration time validation
- Issuer and audience claims validation
- Required scopes/roles validation

Token Refresh: When the access token expires, use the refresh token to obtain a new access token:

```
POST /auth/realms/{realm}/protocol/openid-connect/token
Content-Type: application/x-www-form-urlencoded
```

```
grant_type=refresh_token&
client_id=jul-client&
client_secret={client_secret}&
refresh_token={refresh_token}
```

Token Lifecycle:

- **Access Token Lifetime:** 60 minutes (configurable)
- **Refresh Token Lifetime:** 30 minutes (configurable)
- **Session Timeout:** Configurable per realm
- **Token Revocation:** Supported via logout endpoint

8.3.5 Integration Requirements**Technical Requirements:**

1. **Keycloak Server:** Version 21.0 or higher
2. **TLS/SSL:** All communication must use HTTPS
3. **Network Access:** JUL system must have network access to Keycloak server
4. **Client Configuration:** Proper client registration in Keycloak
5. **Role Mapping:** Define and map user roles for authorization

Security Requirements:

- Secure storage of client credentials
- Token encryption in transit
- Token validation on every API call
- Proper error handling for authentication failures
- Session timeout and automatic logout

Implementation Checklist:

- ☐ Register JUL application as Keycloak client
- ☐ Configure redirect URLs and valid origins
- ☐ Set up user roles and permissions
- ☐ Implement token acquisition flow
- ☐ Implement token refresh mechanism
- ☐ Add authorization header to all API calls
- ☐ Implement proper logout functionality
- ☐ Test token expiration and refresh
- ☐ Implement error handling for auth failures
- ☐ Document authentication flows for developers

End of Interface Control Document

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